

Colibri Hybrid Engine Technical Report

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1 Project Overview

This document outlines the design of a hybrid rocket engine designed to compete in the Spaceport America 10K Hybrid/Liquid category. The engine produces an average thrust of 1500 N over an 8.33 second period. The oxidizer feed system uses the high vapor pressure of Nitrous Oxide at ambient conditions to serve as the pressurization system. The oxidizer tank and combustion chamber will be housed in a 6 in. and 5 in. OD aluminum frame respectively. The entire engine is approximately 1.6 meters in height.

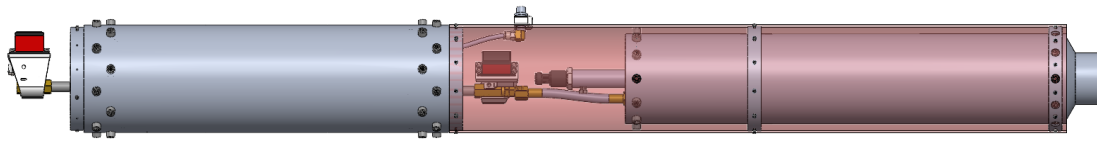


Figure 1: Engine Assembly

Originally our team planned on designing, manufacturing, and testing the hybrid motor. Unfortunately, unforeseen delays in material selection and ordering made it so we did not get the parts in time. We needed a lot of very specific parts, that were hard to find with reasonable shipping times to Mexico. Ultimately, this meant we were not successful in manufacturing the parts in time which prevented us from testing. However, we aim for this paper to serve as a guide for amateur rocketry teams to have a good base for designing their own hybrid motors.

2 Propellant Selection

2.1 Fuel

To achieve the best combination of cost, ease of access, and performance, the team decided to use Paraffin as the fuel for the hybrid engine. Additionally, Paraffin is relatively easy to cast, and it does not present any health hazards when working with it. The team did not see Hydroxyl Terminated Polybutadiene (HTPB) as a viable option despite higher efficiency as compared to Paraffin, because it is not easily obtained in Mexico due to army regulations.

Paraffin has some downsides. In its solid form, it is brittle, and does not have the same mechanical properties of HTPB that allow it to maintain its integrity through a burn. For this reason, Paraffin requires a binder or similar solution to provide reinforcement and prevent premature melting or slumping of the fuel grain. Additionally, paraffin is usually a translucent white material. This is disadvantageous because it allows for light waves produced during combustion to reach the internal portion of the wax, and melt it before it can combust. This contributes to the poor structural integrity of the material. Later in the report, we will present solutions to these problems.

2.2 Oxidizer

Considering the storage and safety concerns of cryogenic oxidizers, Nitrous Oxide was chosen as a safer alternative due to it being non-cryogenic at ambient conditions. At ambient temperatures, Nitrous Oxide has a high vapor-pressure, allowing it to serve as our pressurizing system in place of an inert, back-filling gas. Additionally, Nitrous Oxide is widely available due to its wide commercial applications.

Despite its advantages, Nitrous Oxide can be difficult to work with due to its large change in pressure over different temperatures. While the performance of our engine largely depends on the vapor pressure of Nitrous Oxide directly before combustion, it may be difficult to predict the exact temperature and thus pressure expected during testing. Additionally, while Nitrous Oxide is able to maintain its high pressures over time, it is inevitable that there will be a pressure loss over the course of the burn, making modeling more difficult compared to a back-filled engine.

3 Engine Performance Requirements

3.1 Chamber Pressure

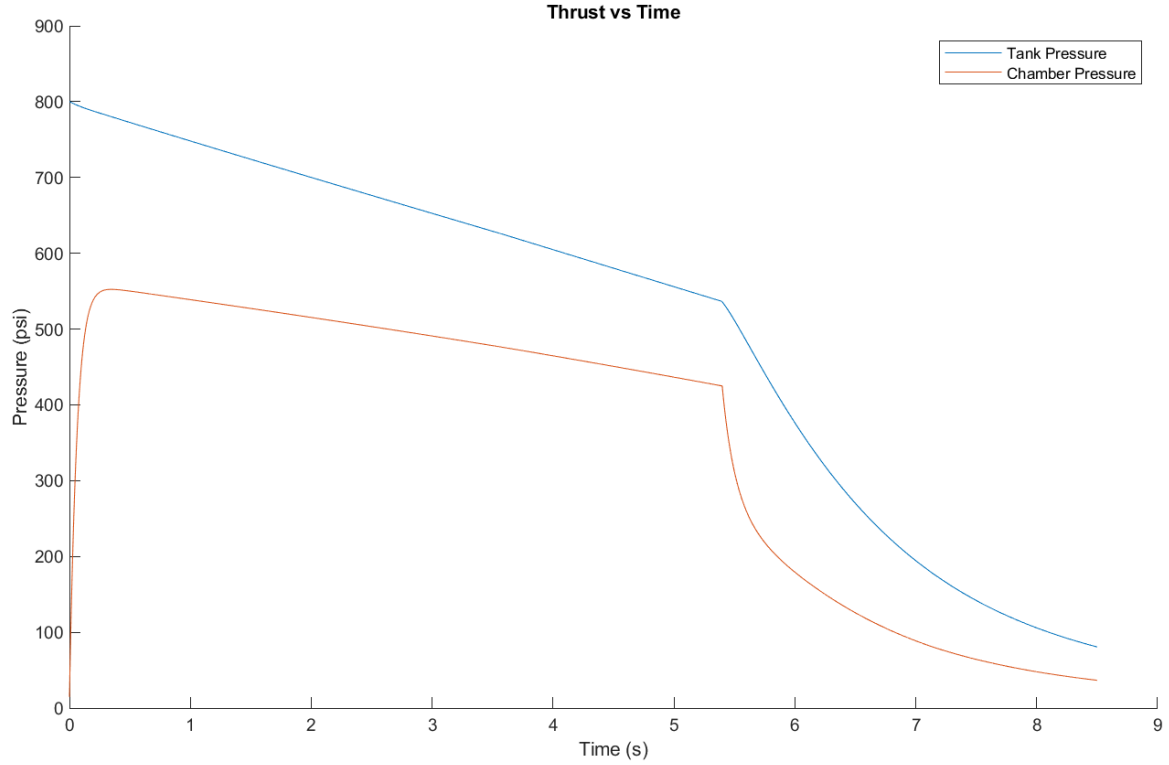


Figure 2: Pressure Graph

The engine is designed to achieve an average chamber pressure of 350 psi. Chamber pressure is directly related to pressure of the oxidizer tank and the pressure drop across the injector during combustion, and is constrained by the self-pressurization characteristics of Nitrous Oxide in our design.

While higher chamber pressures typically correlate to better performance, an average of 350 psi is a typical chamber pressure for engines utilizing the high vapor pressure of Nitrous Oxide and was chosen to prevent the back-flow of fluid as the oxidizer tank pressure decreases with time. The following graph depicts chamber pressure and oxidizer tank pressure over time.

3.2 OF Ratio

Ingredients

Name	Propellant Name	Weight (gr)
	NITROUS OXIDE	4381.00
	PARAFFINIC OIL	674.00
	CARBON BLACK	6.74
		0.00
		0.00
		0.00
		0.00
		0.00
		0.00
		0.00
		0.00
		0.00
		0.00
		0.00
		0.00
		0.00
Total Wt. (grams)		5061.74

Operating Conditions

Temp. of Ingredients (K)

Chamber Pressure (PSI)

Exhaust Pressure (PSI)

☐ Boost Velocity and Nozzle Design

Calculate

Isp*

202.3844

C*

5272.085

Density

7.503158E

Molecular Wt.

25.73111

Chamber CP/CV

1.250333

Chamber Temp.

3282.696

Display Results

Display Nozzle Graphs

Figure 3: ProPep Configuration

An OF ratio of 6.5 was chosen based on propellant analysis using the program ProPep. With Paraffinic Oil and Nitrous Oxide as our propellants, we varied the ratio of oxidizer and fuel to find the highest possible Isp. Operating conditions reflected our desired chamber pressure of 350 psi and typical atmospheric conditions in Mexico. Through our analysis, we found that the a maximum Isp of 202.47 occurred around an OF ratio of 6.5.

3.3 Average Thrust, Burn Time

The engine is designed to produce an average thrust of 1500 N with an 8.33 second burn time. These design parameters were chosen through thrust and flight analysis on the programs OpenRocket and HRAP (Hybrid Rocket Analysis Program). Using the previously defined parameters and estimates of our grain, nozzle, and injector geometries, we found that an average thrust of 1500N with a burn time of around 8 seconds would take us to our desired apogee of 10,000 ft. To account for inevitable thrust inefficiencies, we are designing the engine to fly 10 percent above our desired apogee to approximately 11,000 ft, corresponding to an 8.33 second burn time. After our first static fire, we intend to resize the fuel grain as necessary for maximum efficiency at our desired apogee.

3.4 Mass Flow Rates

Because of our method of pressurizing the oxidizer feed system and the nature of burning fuel grain, it is inevitable that our oxidizer and mass flow rates will vary with time. To continue design of our

hybrid engine, it was important to determine a desired average flow rate to design around. Average propellant, oxidizer, and fuel mass flow rates were calculated using the following formulas:

$$m_p = \frac{F_t}{I_{sp} * g} = \frac{1500}{202.47 * 9.8} = .755 \text{ kg/s}$$

$$m_f = \frac{m_p}{r + 1} = \frac{.758}{6.5 + 1} = .101 \text{ kg/s}$$

$$m_o = m_p - m_f = .755 - .101 = .654 \text{ kg/s}$$

Using our previously defined OF ratio of 6.5, we found our desired oxidizer mass flow and fuel mass flow to be 0.654 kg/s and 0.101 kg/s respectively. While designing other aspects of our hybrid engine, the team ensured that our average mass flow rate always stayed within the calculated range. Using HRAP to get more accurate mass flow measurements, the following graph demonstrates how our mass flow rates vary with time.

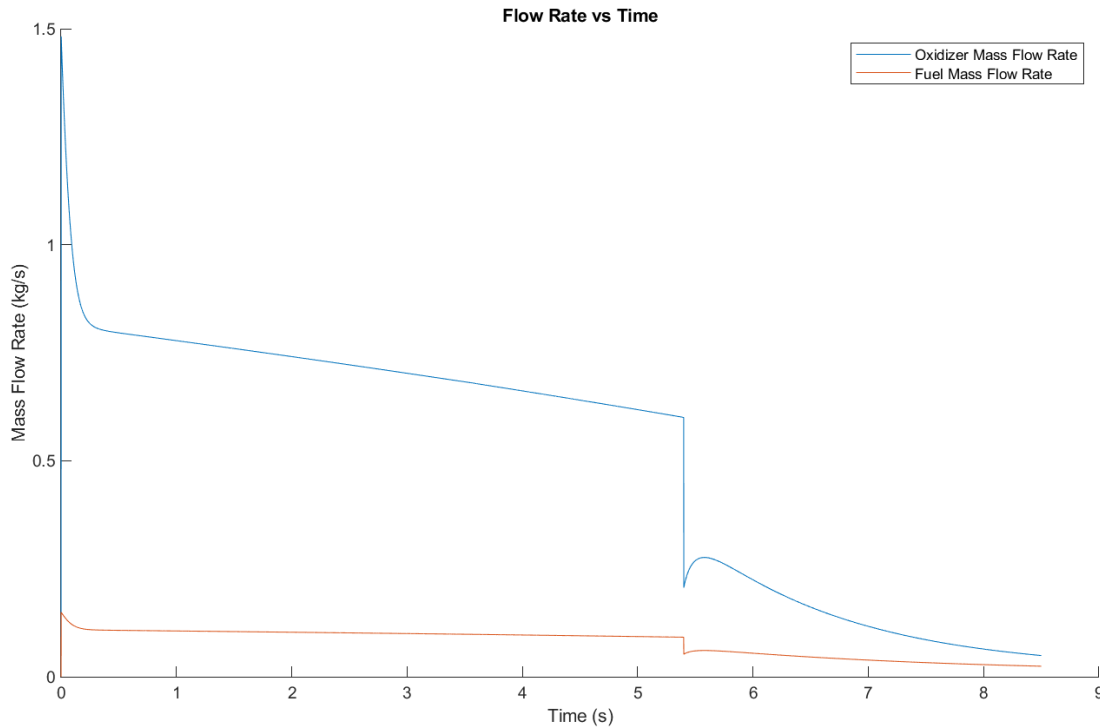


Figure 4: Flow Rates vs Time

Working backwards using an 8.33s burn time, our average mass flow rates, and relevant densities for Paraffin and Nitrous Oxide, we found that we needed approximately 0.839 kg of Paraffin and 5.45 kg of Nitrous Oxide. Using HRAP to get more accurate propellant measurements, we found that we actually need 0.674 kg of Paraffin and 4.23 kg of Nitrous Oxide exactly. The lower required mass can be attributed to the fact that HRAP estimates lower average mass flow rates than the team calculated. This can be attributed to the unpredictability of mass flow rate as Nitrous Oxide's vapor pressure decreases over time.

4 Oxidizer Feed System

The purpose of the oxidizer feed system is introduce oxidizer to the injector where it is fed into the combustion chamber. The assembly is placed on top of the combustion chamber and must be able to safely contain Nitrous Oxide at its highest expected operating pressures, which we expect to be around 800 - 900 psi.

4.1 Oxidizer Feed System Assembly

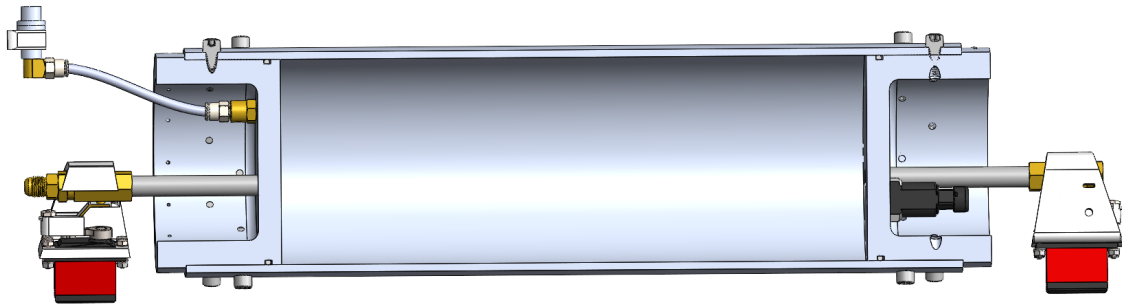


Figure 5: Oxidizer Feed System Cross-Section

As pictured above, the oxidizer feed system consists of the oxidizer tank, bulkheads, quick-disconnect, servo-valve assemblies, pressure transducer, and necessary piping. The total assembly weighs approximately 8.55 kg and is 0.74 meters long. It's important to note that the outer casing of the oxidizer tank will serve as the outer casing of the rocket. While the assembly would benefit aerodynamically from flush screws in this configuration, we determined that the effects were negligible.

4.2 Oxidizer Tank

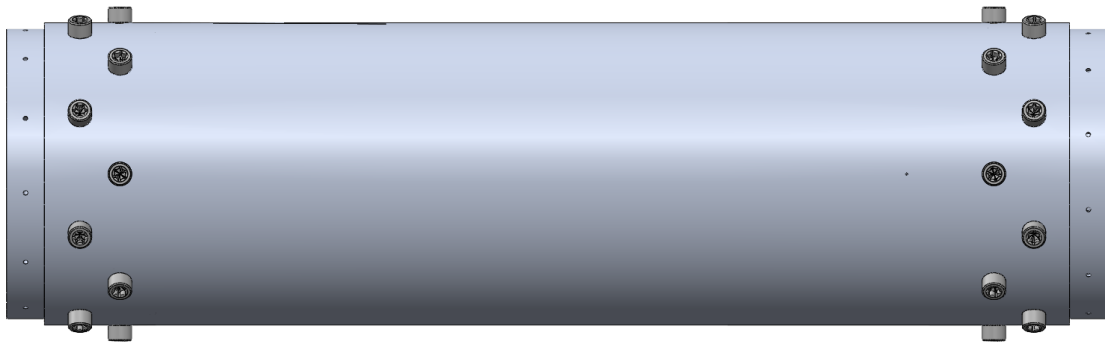


Figure 6: Oxidizer Tank Assembly

The oxidizer tank consists of a 6 inch OD case, 2 bulkheads, and 32 black- oxide alloy steel screws. To achieve our desired burn time of 8.33 seconds, we will need to store 4.23 kg of Nitrous Oxide.

At a temperature of 75°F, Nitrous Oxide has a density of 753.03 kg/m^3 . To have adequate space for our oxidizer at this density, our oxidizer tank will be 6.25 liters in volume and have a height of 21.86 inches. 5 percent of the oxidizer tank will be dedicated to the ullage, leaving ample space for vaporized Nitrous Oxide.

4.2.1 Oxidizer Tank Case

The 20.36 inch oxidizer tank case is machined out of Aluminum 6061-T6 and weighs approximately 6.75 pounds. There are two rows of 8 bolt holes on either side of the case to allow for the fastening to the oxidizer tank bulkheads. The tank case is designed to withstand pressures up to twice of our max expected operating pressure of 960 psi, which corresponds to the vapor pressure of Nitrous Oxide at 90 degrees Fahrenheit.



Figure 7: Oxidizer Tank Case

4.2.2 Static Vent

In order to allow for the flow of Nitrous Oxide into the oxidizer tank case during filling, a static vent will be drilled into the wall of the oxidizer tank. The static vent functionally allows air to escape the tank as Nitrous Oxide is introduced; without the use of a static vent, the oxidizer tank would never fill with oxidizer because the air and Nitrous Oxide would eventually reach equilibrium.

The size of the static vent orifice directly affects the fill time of the oxidizer tank, as determined by the volumetric flow rate of gas through the orifice. Using the "Oxidizer Vent" section of HalfCatSim, we decided on an orifice size of 1.0 mm, which corresponds to a 48 s fill time. We know our tank is full when liquid Nitrous Oxide begins to leak out of the static vent; to achieve our 5 percent desire ullage volume, the static vent will be placed 3.518 inches from the top of the oxidizer tank case.

4.2.3 Stress Calculations

An analysis was conducted of the hoop and axial stress experienced by the oxidizer tank during its highest expected operating pressure. During testing of our hybrid engine, we expect temperatures to be around 75 degrees Fahrenheit at the time of filling. This means the Nitrous Oxide will have a vapor pressure of 799.74 psi. Considering the possibility of temperatures rising above this, stress calculations were conducted assuming a temperature of 90 degrees Fahrenheit, corresponding to a

vapor pressure of 960.46 psi. The following equations were used to calculate hoop and axial stress:

$$\sigma_h = \frac{P_{max} * r_{mean}}{t} = \frac{960.46 * 2.90625}{.1875} = 15,367 \text{ psi}$$

$$\sigma_a = \frac{P_{max} * (D - t)}{4 * t} = \frac{960.46 * (6 - .1875)}{4 * .1875} = 7443 \text{ psi}$$

Using a 6 in. OD aluminum case with a thickness of .1875 in., we found that the oxidizer tank will experience a hoop stress of 15,367 psi and an axial stress of 7443 psi. Aluminum 6061 has a yield strength 35,000 psi, putting our factor of safety at a comfortable 2.27 for hoop stress and 4.70 for axial stress.

4.3 Oxidizer Tank Bulkheads

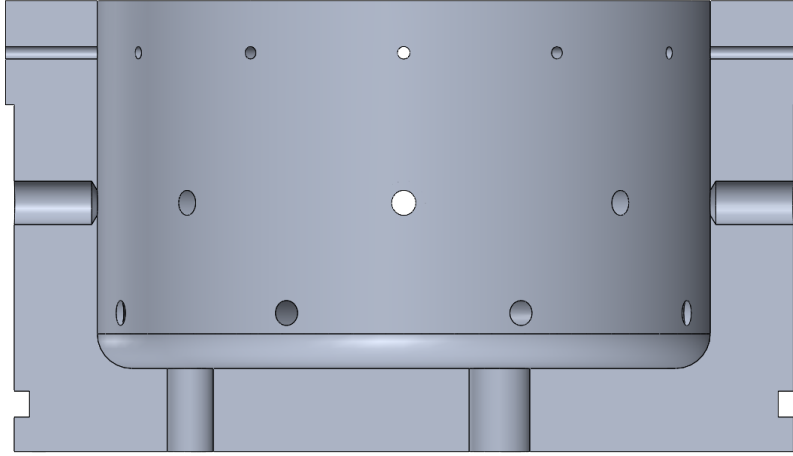


Figure 8: Bulkhead Cross-Section

The purpose of the oxidizer tank bulkheads is to contain the high pressures experienced within the oxidizer tank, while serving as a fastening point to the oxidizer tank case and fiberglass tubes. Each bulkhead weighs about 3.88 pounds and is machined out of the same Aluminum 6061-T6 used for the oxidizer tank case. The bulkhead is 0.60 inches thick which has been proven to achieve our desired factor of safety through SolidWorks FEA simulation. The carved out design of the bulkhead was chosen to achieve the desired factor of safety while minimizing the amount of material used, thus minimizing mass.

The top and bottom bulkheads contain different sized NPT ports for the necessary instrumentation screwed to each. The top bulkhead located on the top of the oxidizer tank contains two 1/4 inch NPT ports for the servo-valve assembly and a pressure transducer. The bottom bulkhead contains a 1/8 inch and 1/4 inch NPT port, for the quick disconnect tube and servo-valve respectively.

Each bulkhead contains a total of 32 bolt holes: 16 for fastening to fiberglass and 16 for fastening to the oxidizer tank. Bolt holes for fastening to the oxidizer tank are .3125 inches in diameter, while the bolt holes for connection to the fiberglass tubes are .089 inches in diameter.

4.3.1 Bulkhead Finite Element Analysis

To confirm that our oxidizer tank bulkheads are able to sustain the high pressures experienced within the oxidizer tank, the team conducted SolidWorks FEA on the component. The simulation was configured with the 32 bolt holes selected as fixed geometries, since these shouldn't move during testing or flight, and a 960 psi load placed on the side of the bulkhead inside of the tank. The following figure depicts the Von-Mises stress experienced by the bulkhead at the maximum expected operating pressure:

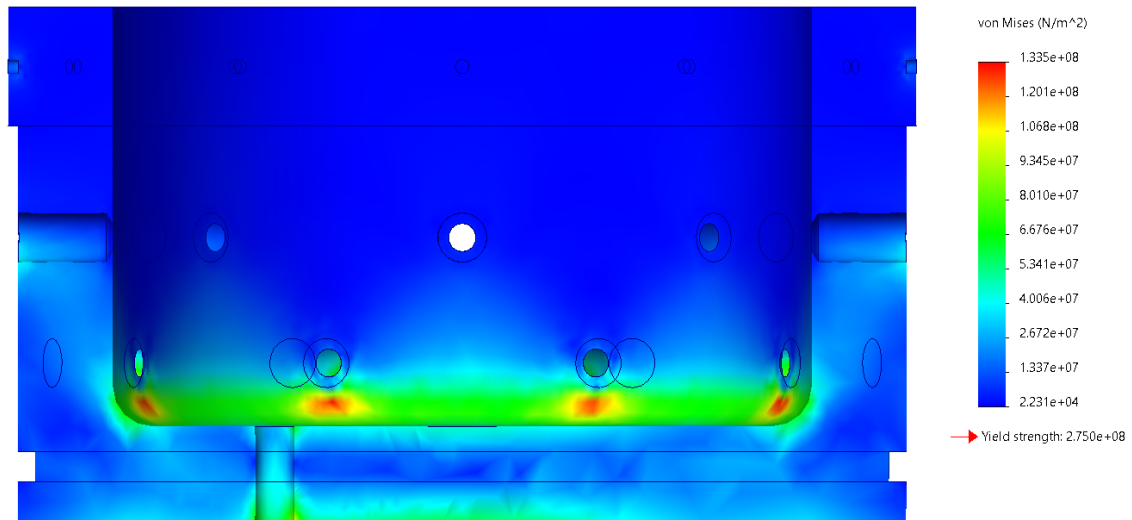


Figure 9: Bulkhead Von-Mises Simulation

As shown in the figure, the maximum Von-Mises stress experienced is less than half the yield strength of Aluminum 6061, meaning that our minimum factor of safety is above 2. To confirm our factor of safeties, an FOS simulation was also conducted using SolidWorks:

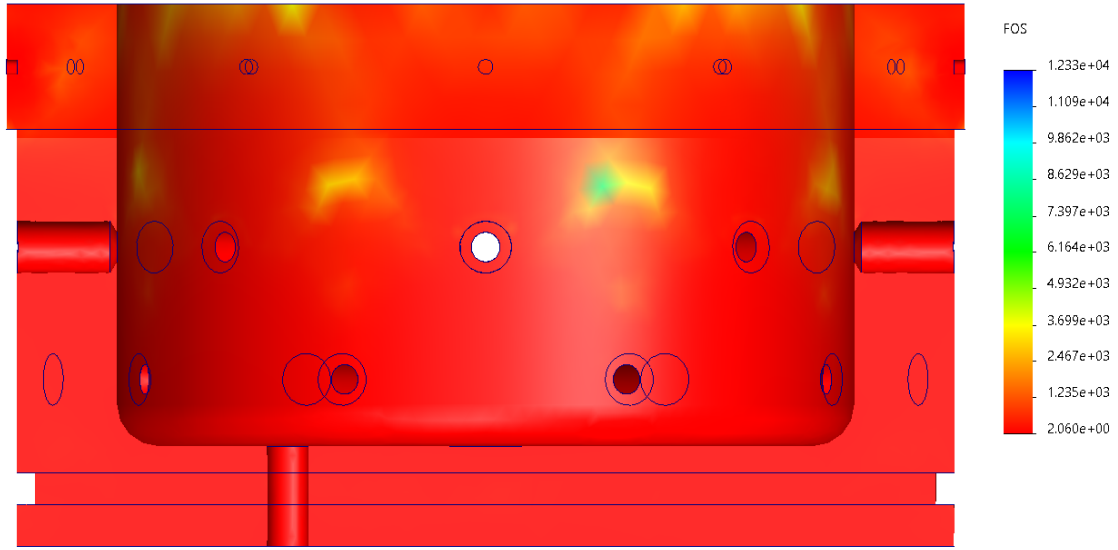


Figure 10: Bulkhead FOS Simulation

This simulation confirms that our minimum factor of safety is above 2: at 2.06 to be precise. Since our FOS is above 2, our bulkheads should be more than strong enough to contain the high pressures expected inside of the oxidizer tank.

4.3.2 Bolt Calculations

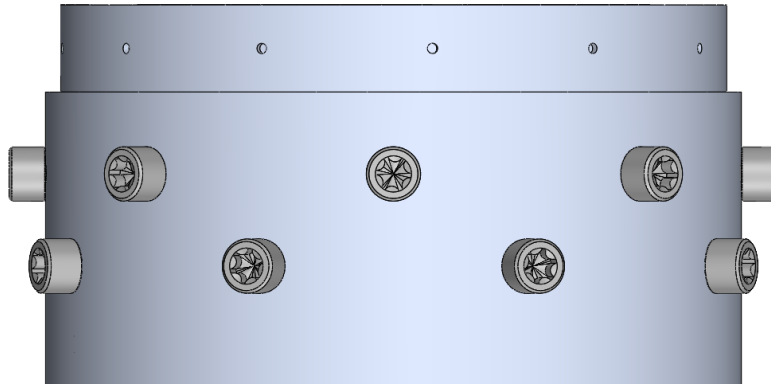


Figure 11: Bulkhead Fastening

To secure the oxidizer tank to its bulkheads, two rows of eight 5/16-18 Torx screws will be used on each side. To ensure that the screws would be able to withstand the high pressures contained within the oxidizer tank, we conducted multiple bolt calculations to ensure that the stresses experienced

fell within our desired factor of safety of 2. We began with calculating the force on each bolt at max operating pressure:

$$F_{bolt} = \frac{\pi * D_{i,casing}^2 * MEOP}{4 * N} = \frac{\pi * 5.625^2 * 960}{4 * 16} = 1491.56 \text{ lbf}$$

Using this force, we are able to calculate the shear, tear-out, and bearing stress experienced by each bolt:

$$\sigma_{shear} = \frac{4 * F_{bolt}}{\pi * d_{bolt,minor}^2} = \frac{4 * 1491.56}{\pi * .252^2} = 29895 \text{ psi}$$

$$\sigma_{tear-out} = \frac{F_{bolt}}{E_{min} * 2 * t} = \frac{1491.56}{.55 * 2 * .1875} = 7232 \text{ psi}$$

$$\sigma_{bearing} = \frac{F_{bolt}}{D_{bolt,major} * t} = \frac{1491.56}{.313 * .1875} = 25145 \text{ psi}$$

Using the ultimate tensile strength of steel bolts (180,000 psi), the shear strength of aluminum 6061 (30,000 psi), and the bearing strength of aluminum 6061 (56,000 psi), we are able to confirm that all our factor of safeties are greater than 2. We find the factory of safety by dividing the each by their relevant calculated bolt stress.

Additionally, we calculate the tensile stress experienced between the fastener holes to ensure it is not stretched beyond its breaking point:

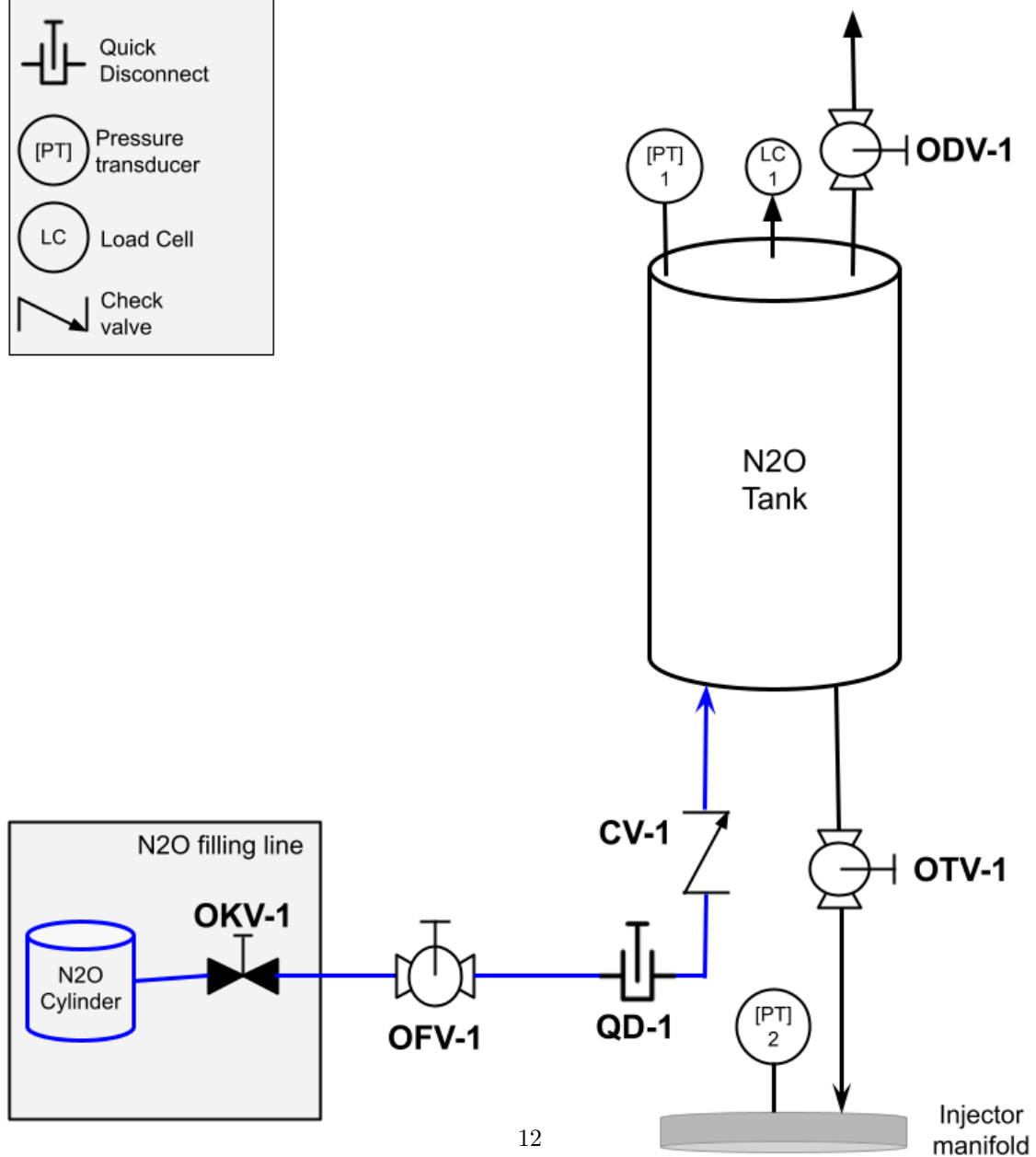
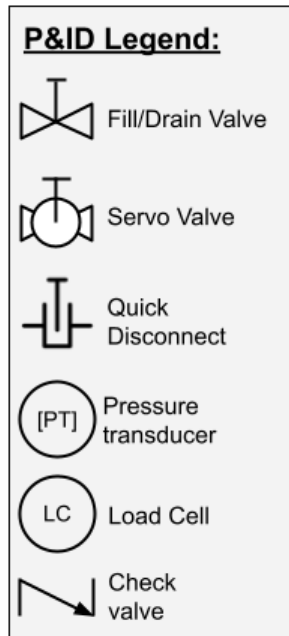
$$\sigma_{tensile} = \frac{\frac{\pi}{4} * D_{case,inner}^2 * P_{max}}{((D_{case,outer} - t) * \pi - (N * d_{bolt,major})) * t}$$

$$\sigma_{tensile} = \frac{4 * 5.625^2 * 960}{((6 - .1875) * \pi - (16 * .313)) * .1875} = 9601 \text{ psi}$$

Using the yield tensile strength of aluminum 6061 (38,000 psi), we are able to confirm that our factor of safety is also above our desired number of 2.

4.4 Piping and Instrumentation

4.4.1 Piping and Instrumentation Diagram



4.4.2 Servo-Actuated Ball Valve (FV-1 and DV-1)

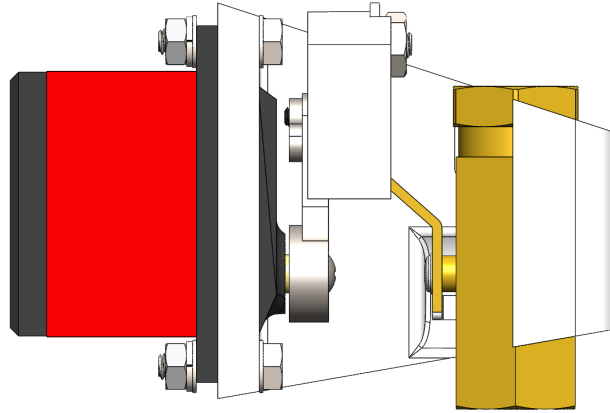


Figure 12: Servo-Valve Assembly

The use of a servo-actuated ball valve is a simple solution for remotely filling our oxidizer tank. Compared to the typical solenoid valve, a servo-actuated ball valve is significantly cheaper and simpler. The valve works by connecting a servo motor and manual ball valve to a 3D printed casing. When the motor is connected to a receiver, the motor can be actuated to open the manual ball valve with a wireless transmitter.

This ball valve will be used as both our feed valve and drain valve. The feed valve is located below the oxidizer tank and functionally allows for the flow of oxidizer to the injector where it is fed into the combustion chamber. The drain valve is located below the oxidizer tank and is used in the scenario where we need to quickly alleviate pressure in the tank. Considering that our highest expected operating pressure is 960 psi, the drain valve will be actuated if the tank reaches a pressure above that.

Lastly, this ball valve will also be used on the Nitrous Oxide fill line. When filling our oxidizer tank, it is important for personnel to be 200 ft. away because of the dangers of a student manufactured pressurized vessel. Using this variation of a ball valve allows us to control the flow of oxidizer from the Nitrous Oxide cylinder into our oxidizer tank from up to 200 ft. away.

4.4.3 Quick Disconnect (QD-1)

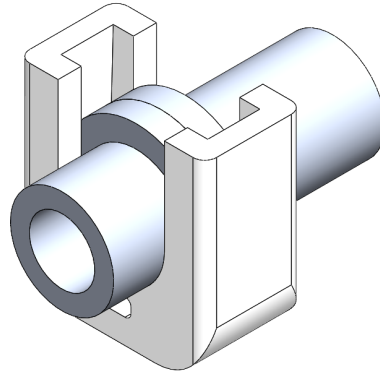


Figure 13: Quick Disconnect Assembly

A flange-clip quick disconnect will be used as the connection between our Nitrous Oxide dewar and oxidizer tank. The purpose of the quick disconnect is to sever the rocket from the fill line when the rocket launches. The mechanism is a custom machined part with a male and female component. When pressurized fluid is flowing through the QD, the linear force of the fluid will attempt to separate the male and female components. A clip will prevent this separation during filling, and will come undone once the rocket launches off the launch pad. The graphic below depicts how the QD will operate during launch.

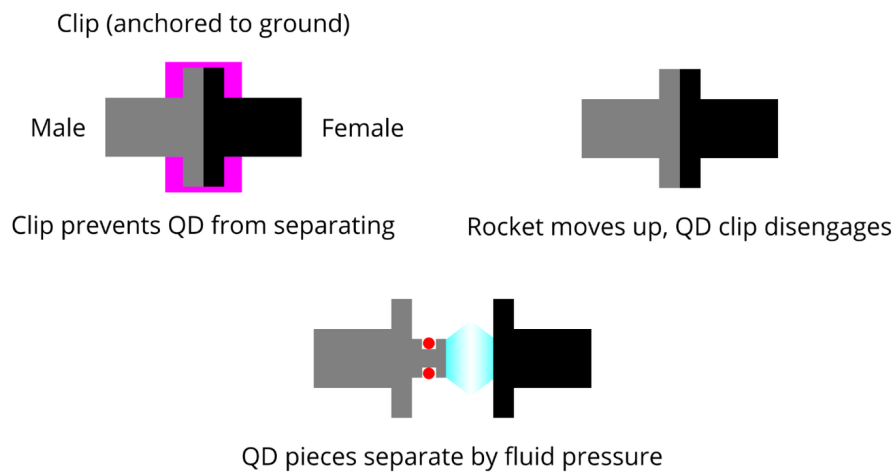


Figure 14: Quick Disconnect Graphic

4.4.4 Check Valve (CV-1)

To prevent the flow of Nitrous Oxide out of the fill line after launch, a check valve will be used between the oxidizer tank and fill line. The check valve only allows flow one way: in the direction of the oxidizer tank. Through the use of a check valve, we are able to allow fluid to enter the oxidizer tank while preventing fluid from escaping after launch.

4.4.5 Pressure Transducer (PT-1)

The oxidizer tank pressure transducer is pivotal for knowing the vapor pressure of our oxidizer at any given moment. Given how much Nitrous Oxide's vapor pressure varies depending on temperature, the live feed pressure data allows us to know if our tank is reaching unsafe conditions. In the event that the tank pressure reads over 1000 psi, we will actuate the drain valve to quickly alleviate all pressure from the tank and drain all oxidizer.

4.4.6 Tubing

Considering the need for non-rigid piping, the oxidizer feed system will take advantage of nylon tubing and a braided steel hose for the quick disconnect assembly and feed valve respectively. While nylon tubing is not rated for high pressures, it should be satisfactory for the significantly lower pressures expected in the Nitrous Oxide cylinder and is flexible enough to easily work with. A 1/4" 37° JIC stainless steel hose is used between the oxidizer tank and injector. This tube is rated up to 3000 psi, more than enough for the highest pressures expected in our oxidizer tank while also offering flexibility.

4.5 O-Ring Sizing

O-Rings must be used on the bulkheads of the oxidizer tank and the male-end of the quick disconnect to prevent the leakage of Nitrous Oxide. Leakage of Nitrous Oxide is dangerous due to its reactivity as an oxidizer. All O-rings on the oxidizer feed system will be made of low-temperature silicon, to handle the low temperatures of liquid Nitrous Oxide that we're filling with. The following table describes the O-ring specifications for the relevant components in the oxidizer feed system:

Component	Bulkheads	QD	Tolerances
O-Ring Series	2-253	2-007	N/A
Groove Depth	5.403"	0.181 "	+ 000 "
Plug Diameter	5.622"	0.279 "	+ 000 001 "
Groove Width	0.187"	0.093 "	+ 005 "

5 Combustion Chamber Assembly

5.1 Combustion Chamber Overview

The combustion chamber is the place where all the action goes down. Here, our Nitrous Oxide oxidizer atomizes, distributes itself, and once heat is introduced, it reacts with the Paraffin fuel to create energy that we use as thrust to propel our rocket forward. All of this means we have to deal with high pressures, and hot temperatures. We will address both of these issues in the sections below.

This is the current assembly, integrating all the components we will mention in this section.



Figure 15: Combustion Chamber Assembly, Cross Sectional View

5.2 Igniter

To begin combustion in our engine, our chemical reactants need heat. To get this heat, we will use a solid propellant cartridge igniter. The solid propellant cartridge will be an "A" class Estes motor (approximately 0.5" OD, 0.75" long), which is easily acquired here in Mexico. The igniter is responsible for decomposing the Nitrous Oxide into nitrogen and oxygen, which then reacts with the fuel and provides stable combustion. To ensure that the igniter does not cause deformities in our fuel grain, it will be placed in the middle of the injector assembly. Additionally, since the igniter hole will be directly exposed to the combustion chamber (and thus the pressure inside of the chamber), a pressure transducer will be placed on top of the igniter cartridge to measure chamber pressure throughout the burn.

The igniter will be connected to the injector using of a 1/4" Steel NPT fitting seals against the threads, and a machined aluminum cartridge that the pressure transducer will be threaded into. Teflon tape will be used on all pressure bearing threaded connections to ensure a good seal. An e-match will be used to create heat from electricity, and light the Estes motor. To hold the e-match in place, it will be inserted into a tapered 1/8" NPT steel plug, that will thread into the igniter cartridge just above the top of the Estes motor. The e-match will be held in place by high temperature epoxy. To prevent the epoxy from popping out due to high pressure, the NPT plug will have a tapered hole design, using progressively smaller drill bits to ensure the epoxy has material to push against and prevent it from popping out.

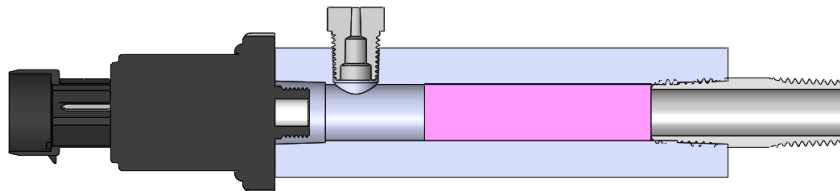


Figure 16: Igniter, Cross Sectional View

The internal diameter of the igniter cartridge is constrained by the drill size needed for a 1/4" NPT tap. Since this requires a 7/16" drill bit, but the outer diameter of the Estes motor is 1/2", we would need to shave down some of the outside of the motor so it will fit. Additionally, to protect the pressure transducer from the high temperatures from ignition and combustion within the combustion chamber, the open end will be filled with high temperature resistant grease that will absorb heat while still deforming the diaphragm to allow for pressure readings.

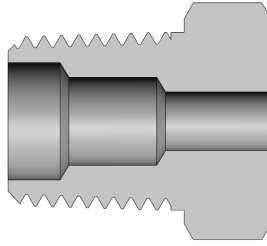


Figure 17: Steel Plug Taper Design

Pressure from combustion will come through the igniter chamber, and thus from this perspective will be exerted on the igniter plug from the left. Since the drilled hole becomes progressively smaller as you move rightward, there is support material to prevent the epoxy holding the e-match from simply popping out. The drill diameter reduces as follows: $1/4''$, $3/16''$, $1/8''$. Each will increase $1/3$ of the total height of the steel plug beyond the size before it.

The e-match used in the igniter will be an electronic igniter match, commonly used to light fireworks or model rockets. The e-match ignites once a current is passed through the two ends, which goes to a heat generating bridge wire coated in a pyrotechnic material that will ignite, and light the Estes motor.

5.3 Fuel Grain

The fuel will use a Bates geometry. A bates grain uses a cylindrical mandrel to create the desired port area. Despite boosted performance from more complex geometries such as a star grain, wagon wheel, or dog-bone, these are considerably harder to cast, and present an increased danger of the nozzle becoming clogged with chunks of unburned fuel since there are thin, delicate parts associated with each. Additionally, HRAP, our simulation software, only took in parameters for a Bates geometry, so if we wanted to use a more complex grain, we would need to find or make a new simulation software. The external diameter of the grain is $4.25''$, this takes into account the $1/8''$ thickness of the ablative liner within the combustion chamber, and the $1/8''$ thickness of the phenolic casting tube.

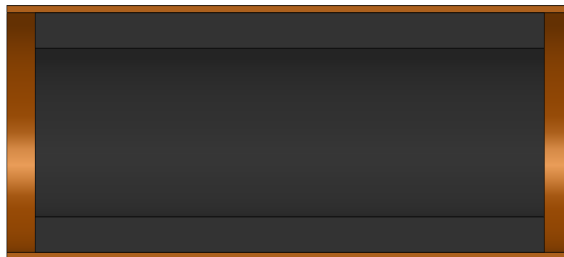


Figure 18: Fuel Grain

You may notice that there is a small section of phenolic casting tube on either side of the paraffin fuel grain. This $1/2''$ section on either side is to allow for the casting caps to be securely attach and

prevent paraffin from leaking out during spin casting. These sections will not contain paraffin, and so will form part of the pre and post combustion chambers.

5.3.1 Ballistic Coefficients

Typical models for the regression rate of propellant grains utilize an exponential model with a regression rate coefficient (a , in units of m/s), and a regression rate exponent (n , unit-less). Since our team does not currently have the capacity to experimentally determine these figures, we found resources online that compiled the average values of a and n in a Paraffin - Nitrous Oxide propellant mixture. The values came out to be 0.1435 m/s, and $.5275$ for a , and n , respectively.

Fuel	Oxidiser	a	n	Reference
HTPB	GOX	0.025×10^{-3}	0.65	Karabeyoglu et al. (2002)
HTPB	GOX	0.0304×10^{-3}	0.681	Sutton and Biblarz (2001)
HTPB	GOX	0.087×10^{-3}	0.53	George et al. (2001)
HTPB	GOX	0.077×10^{-3}	0.53	George et al. (2001)
HTPB	N ₂ O	0.094×10^{-3}	0.325	Lohner et al. (2006)
Paraffin wax	GOX	0.091×10^{-3}	0.69	Karabeyoglu et al. (2002)
Paraffin wax	LOX	0.117×10^{-3}	0.62	Karabeyoglu et al. (2003)
Paraffin wax	H ₂ O ₂	0.0344×10^{-3}	0.959	Brown and Lydon (2005)
Paraffin wax	N ₂ O	0.132×10^{-3}	0.555	Grosse (2009)
Paraffin wax	N ₂ O	0.155×10^{-3}	0.5	McCormick et al. (2005)
HDPE	N ₂ O	0.0462×10^{-3}	0.352	Lohner et al. (2006)
PMMA	N ₂ O	0.0466×10^{-3}	0.377	Lohner et al. (2006)

Figure 19: Experimental Values for ballistic coefficients a , n

Notably, these coefficients are for a pure Paraffin - Nitrous Oxide propellant mixture. In our case, the armored grain, and the incorporation of Carbon Black will lead to changes in these experimental coefficients. However, we will base our calculations off of these numbers since it is not possible to determine them without testing.

5.3.2 Regression Rate Equation

The regression rate of the solid fuel is affected by a variety of parameters including the fuel grain geometry and size, the chemical composition of the fuel and oxidizer, and the internal flow and pressure of gases within the combustion chamber. It is difficult to accurately compute the values of a and n numerically, so experimentally obtained values are used instead. In the case of a Paraffin - Nitrous Oxide fuel mixture, the values are shown above. the regression rate equation uses these numbers to model the rate at which the fuel grain burns, in the following equation.

$$\frac{dn}{dt} = a * \left(\frac{m'_{ox}}{A}\right)^n$$

Where:

A = port area

m'_{ox} = oxidizer mass flow rate

The following derivations for the geometry of the Paraffin fuel grain are based off this regression rate equation and the burn rate coefficients a , and n .

5.3.3 Initial Port Radius

The initial port radius depends on the regression rate of fuel and the amount of desired leftover fuel grain after the burn. Considering the relatively long burn time of 8.33 s, the team found it important to provide insulation to our combustion chamber casing during our burn. We thus decided to have around 0.15 inches of Paraffin fuel leftover after burnout. To estimate the initial port radius of the Paraffin fuel, the team utilized the following formula:

$$r_0 = (-(2n + 1) * t_f * a * (\frac{m'_{ox}}{\pi})^n + r_{fin}^{2n+1})^{\frac{1}{2n+1}}$$

$$r_0 = (-(2 * .5275 + 1) * 8.33 * .1435 * (\frac{1.442}{\pi})^{.5275} + 1.975^{2 * .5275 + 1})^{\frac{1}{2 * .5275 + 1}}$$

This resulted in an estimated initial radius of, $r_0 = 1.54$ in. However, it is important to note that the regression rate decreases over time due to a changing mass flow of oxidizer. This means that the calculated port radius will be larger than needed, and requires an examination of the regression rate over time to get an accurate port radius. To do this, the team plugged the estimated port radius into HRAP and adjusted values until our desired port diameter at burnout was achieved. Since HRAP analyzes the regression rate over time, we got a more accurate initial port radius of 3.0 in.

5.3.4 Fuel Grain Length

The fuel grain length directly affects the oxidizer/fuel ratio of our burn and is where we achieve our desired OF ratio of 6.5. Grain length depends on the initial port radius found earlier. To estimate the grain length needed, our team used the following formula:

$$L = \frac{m'_{fuel} * t}{\pi * (r_{fin}^2 - r_0^2) * \rho} = \frac{.222 * 8.33}{\pi * (1.975^2 - 3.0^2) * .0325}$$

This resulted in an estimated grain length of 11.73 in. However, the calculation of grain length has a similar problem to that of the initial port radius. Since both our oxidizer and fuel mass flow rates are changing, our OF ratio will vary significantly from our desired ratio of 6.5. To reflect the changing the OF ratio, we plugged our calculated value into HRAP and adjusted the grain length until we achieved an average OF ratio of 6.5. Using this method, we determined the optimal grain length to be 9.0 in.

5.3.5 Grain Armor

In order to enhance the regression rate of the propellant, and increase the stiffness of the wax to minimize slumping as it heats up from the combustion process, it will be reinforced with an ABS printed structure. These benefits are explained by the "uneven and irregular texture of the burning surface that promotes turbulence and convective heat transfer" (Bisin, Paravan 2023). Research suggests the ideal shape for this supporting structure is a uniform gyroid pattern. The gyroid

infill structure will be generated using a 3D printing slicing software at a 10% by density infill. Experimentation by Bisin and Paravan with 10% and 15% gyroid infill indicate a higher regression rate for the lower density infill, as a natural result of the higher amount of paraffin present.

For accurate and consistent modeling of the gyroid pattern, we used Ultimaker Cura 5.6 Slicer. This is the cross section of the sliced grain armor in the slicer.

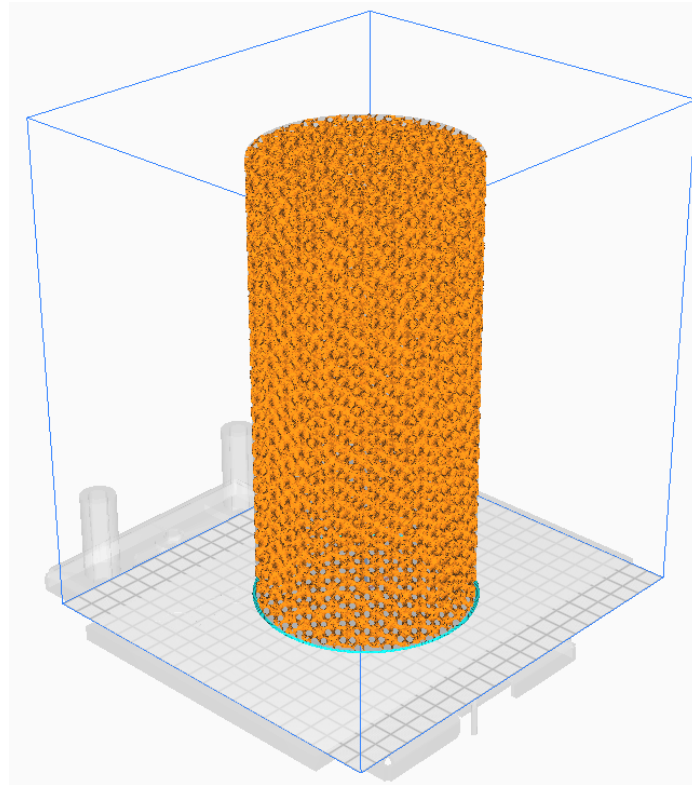


Figure 20: Ablative Liner, Isometric View

An alternative the team explored was using two part expanding Polyurethane Foam (PUF) to act as a binder and provide some stiffness throughout the burn. This was based off a recommendation from a Reddit user who helped formulate the casting procedure. However, we were not entirely sure how we would create a grain to the size we wanted, since the only polyurethane foam we had access to was a) very expensive and b) expanded nearly 16x its original volume. We also found conflicting accounts of just how much we should add, from 30% by mass to 50% by mass. Due to these uncertainties, we decided to go with the armored fuel grain which was easier to manufacture while providing similar benefits.

5.3.6 Carbon Black

As mentioned in the beginning of this report, paraffin wax's translucent white color presents an issue in the form of premature melting of the paraffin that leads to slumping of the fuel grain. A common

solution used in paraffin - Nitrous Oxide engines is the addition of carbon black, a fine black powder that is the product of carbon products combusting incompletely.

Our fuel will use a 1% Carbon Black by mass, to achieve a black color for the wax grain, while not taking away too much from the intended performance. This is reflected in the ProPEP configuration.



Figure 21: Carbon Black powder

5.3.7 Casting Procedure

The majority of the casting hardware will be printed from ABS filament on a Flashforge Adventurer 4 3D printer. This allows for simply designed casting hardware that is easy to assemble, and avoids manufacturing time and material costs from alternative solutions. The only non-ABS component will be the casting shell (outer cylindrical wall) which will be made of a phenolic resin - linen composite.

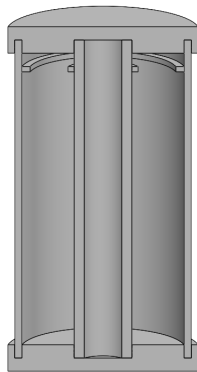


Figure 22: Casting Hardware

The top and bottom casting caps are there to hold the wax in place, and they serve as the attachment point to the chuck on the lathe where the entire assembly will be spin cast. The mandrel in the center is sized according to the initial port radius calculated earlier. The thin ring at the top is removable and helps to center the mandrel when the top cap is not in place. Of these components,

the only one that should permanently stick to the wax is the outer casting tube. In the graphic above, this is made of ABS, since this is a prototype to test out the casting procedure before casting in the phenolic tube.

Since the paraffin wax needs to be heated up to cast, it is important to take into account the heat properties of the ABS casting hardware. ABS has a glass transition temperature (the point at which the material begins to soften) of around 105°C, and no true melting point since it is an amorphous material. We want to stay far below the melting temperature of ABS, and even below the glass transition point, to avoid any abnormalities in the casting hardware that would give us irregular or unexpected geometries.

The granulated powder Paraffin wax we are utilizing is sourced from a local supplier in Mexico City, and has a melting temperature between 40°C and 50°C, or between 50°C and 60°C for soft and hard waxes, respectively. Since the product specification does not list whether the wax is hard or soft, we will test to determine this. The wax will always be melted at the lowest possible temperature, since this avoids bubbling that could introduce air pockets and compromise the integrity of the wax.

Before a full scale cast, the team will conduct a series of smaller casts to refine the casting process and ensure larger casts are successful. The general procedure will be as follows:

1. Assemble the casting assembly. This includes the top cap, bottom cap, mandrel, and the outside mold. All components should fit snugly against each other, but not to the point that they are excessively difficult to assemble or pull apart.
2. The interior surfaces of the caps and mandrel should be very liberally sprayed with mold release, or a similar lubricant that will ensure we can pull the assembly apart when the wax cools. The interior of the phenolic tube should be kept free of mold release since the wax needs to adhere to the tube so it does not slip during combustion.
3. Remove one of the caps, ensuring that the mandrel does not move from the center position. Place the 3D printed gyroid structure in the void where it will be filled with paraffin.
4. Melt the necessary amount of Paraffin wax powder (+10% to account for heat shrinkage during cooling) using a heat source with a high level of control. Take into account the melting temperature of our specific wax powder, from Manis Chemicals. Be careful to melt the wax at as close to that melting temperature as possible, as this helps prevent air bubbles or cracks from forming. These cracks could be detrimental if the wax cracks off instead of melting, and clogs the nozzle leading to a build up of pressure.
5. Add the 1% paraffin powder by weight. Incorporate slowly, but mixing thoroughly so it evenly distributes among the Paraffin mixture. Be very careful to introduce as little air as possible throughout this process.
6. Slowly fill the mold to the specified point, which should be just above the dimension of the actual final mold, to account for shrinkage.
7. Replace the cap that was removed earlier, and ensure there is a good seal.
8. Place the casting assembly on the lathe, at a relatively low spin speed (around 500 RPM, depending on the lathe).
9. Once the wax has been spinning for long enough to cool entirely, carefully remove the assembly.
10. Remove the end caps, and mandrel with extra care that they do not take chunks of wax with them as they are removed.

11. Visually inspect for noticeable cracks, air bubbles, or similar imperfections.
12. Weigh and size the wax, and cut it if necessary. Use a thin saw blade, and cut slowly.

5.4 Thermal Insulation

The bulk of this rocket is constructed of Aluminum 6061-T6, an alloy with a melting temperature of approximately 925° K. Our propellant mixture, on the other hand, burns at a temperature closer to 3240°. If we were to conduct tests or a launch disregarding this, the combustion is almost guaranteed to burn through the chamber, destroying the hardware, and potentially putting people in danger. To avoid these risks, the motor uses insulating and ablative materials that absorb the heat from propellant combustion, and keep the metal below its melting temperature.

5.4.1 Ablative Liner

The main safeguard between the combustion process and the aluminum case is a liner made of an ablative phenolic resin - silica cloth composite material. The ablative property of this material means it essentially vaporizes away when exposed to high temperatures. These vapors carry heat with them, keeping the rest of the motor at a reasonable temperature. If an ablative successfully fulfills its purpose,

This motor will use an ablative liner with a thickness of 0.125 in., and an outer diameter that corresponds to the inner diameter of the case (0.125 in.). The liner extends the length of the combustion chamber, from the nozzle to the injector, and seals against the nozzle. If the liner were to seal against the liner and the case, the liner would essentially become a pressure vessel itself, rather than the aluminum case. Since the phenolic liner is a composite, an-isotropic material with stress properties that are not readily known, we can not determine whether this would be able to hold the pressure of combustion.

Conceivably, the hot gases from combustion could escape through the top of the liner near the injector. However, since we need one free end of liner that is not sealed, and this end will mostly be filled with cool Nitrous Oxide in the pre-combustion chamber, heat escape is not likely to happen. Additionally, previously tested solid motors where there was no seal against the top of the liner did not see failure of the phenolic liner.

Here are the relevant dimensions, along with an isometric view of the ablative liner:

Mass: 1.76 lbs

Length: 20.687 in.

OD: 4.75 in.

Thickness: 0.125 in.

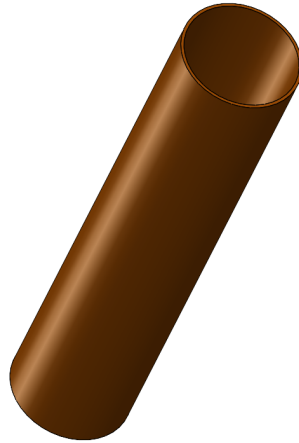


Figure 23: Ablative Liner, Isometric View

5.4.2 Paraffin Fuel

As an additional safeguard from the high combustion temperatures, it is suggested to have some un-burned fuel left over, to act as further insulation. Paraffin by itself is an excellent insulator, and should provide a good amount of thermal insulation in the fuel grain location. Our team decided to have 0.1" of Paraffin fuel left at the end of our burn.

5.5 Pre & Post Combustion Chambers

The pre and post combustion chambers are sections of empty space before and after the fuel grain that each serve a purpose for the successful combustion of the propellant. The material we chose for these components was the same phenolic tubing, selected for its excellent heat properties and its ability to maintain its structure through prolonged heat exposure. This is important because the phenolic tube where the fuel grain is cast needs to be held in place so the fuel grain doesn't slide. One solution we considered was a high temperature epoxy or glue, but there were no available options with melting temperatures close to the expected combustion chamber. The team decided a mechanical connection would be best, specifically one where the pre and post combustion chambers hold the fuel grain in place, preventing it from sliding.

5.5.1 Pre-Combustion Chamber

The pre combustion chamber is situated on the forward end of the rocket, just above the fuel grain. Beyond mechanically attaching to the fuel grain and providing additional thermal insulation, this component helps achieve an efficient combustion. Namely, it provides space that allows for the Oxidizer to reach the required residence time, and it allows the Oxidizer to distribute itself evenly across the area of the combustion chamber. Without this space, the oxidizer particles may be too large which would lead to drops in I_{sp} and thus lower efficiency. A typical approximation for the length of the pre-combustion chamber is $0.5 \times$ the outside of the diameter of the case. This corresponds to 2.5". Since our team is not entirely sure of the combustion characteristics of our propellant, we need to have some space for the fuel grain to lengthen, without having to re-machine

the entire combustion chamber case. For this reason, there is an additional 1.5" of length, for a total length of 4". The CAD and relevant dimensions are shown below. ¹

Length = 3.5 in.
OD = 4.5 in.
Thickness = 0.125 in.
Mass: 0.28 lbs.

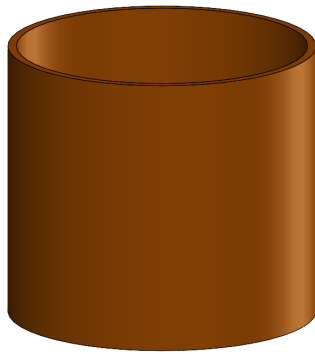


Figure 24: Pre Combustion chamber, isometric

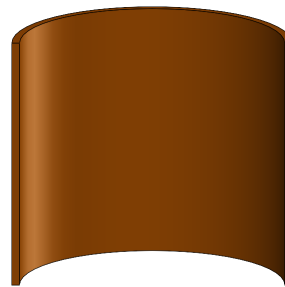


Figure 25: Pre Combustion Chamber, cross section

5.5.2 Post-Combustion Chamber

The post combustion chamber is located across the fuel grain from pre combustion chamber, near the nozzle. This serves the same aforementioned purposes of providing thermal insulation and acting as a mechanical attachment for the fuel grain. Besides this, the post combustion chamber allows for the fuel and oxidizer to properly mix and combust fully. Without this component, it is more likely that there would be unburned fuel / oxidizer which would limit the efficiency of the motor. To dimension this part, a common approximation is to have 1 x the outer diameter of the case, which in this case is 5". As mentioned before, due to uncertainty in the burn characteristics of our propellant mixture we are including additional space for fuel grain expansion, which here will be 2". Thus, the total length of the post combustion chamber is 7".²

Relevant Dimensions:
Length: 6.5 in.
OD: 4.5 in.
Thickness 0.125 in.
Mass: 0.54 lbs

¹This includes 0.5" that is part of the tube holding the paraffin wax

²This includes 0.5" that is part of the tube holding the paraffin wax.

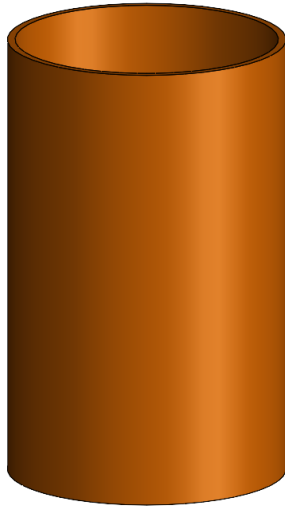


Figure 26: Post Combustion chamber, isometric

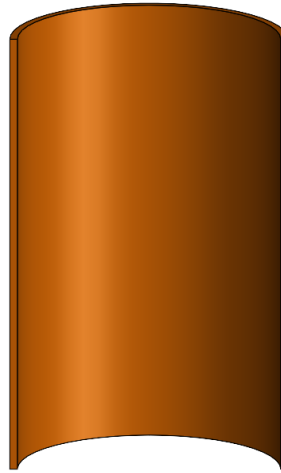


Figure 27: Post Combustion Chamber, cross section

5.6 Nozzle

The nozzle is an integral component in rocket design, as it allows for the thermal energy released from burning propellant to be efficiently converted into kinetic energy that produces thrust and can propel the rocket forward. It is critical that exhaust gases exit only through the throat, as exhaust from any other location could cause the rocket to significantly drift from the intended vertical ascent.

A nozzle design typically consists of an insert (typically graphite, copper, etc.) that contains the throat and some/all of the converging, and diverging section. In our case, the material of choice is graphite, and it will contain the entire converging and diverging sections. This is a very brittle, and porous material. Both of these characteristics require special care in handling, manufacturing, and storing the nozzle. Graphite brittle nature means it is very susceptible to cracks. If you ever drop the graphite nozzle, you can almost guarantee that it has some sort of cracking, and it can no longer be used. If the graphite nozzle ever becomes wet, you need to dry it in a kiln to evaporate all the water that it soaked up, or you risk cracking it when it heats up during a hot fire.

The second part of a nozzle is the nozzle carrier. This serves to hold the nozzle in place, and attach it securely to the rest of the motor. As is the case for the most of our hardware, this is made of Aluminum 6061-T6. This is a structural metal that is relatively inexpensive, and easy to machine. These are all great characteristics since it will have to hold up to the high pressures from combustion, and will need threaded holes to bolt onto the case. The interface between the graphite insert and the nozzle carrier is typically an angled section where the graphite insert will press against when pressure begins to build up. To prevent exhaust escape from this section, we connect the two components using a room temperature vulcanizing silicone. In our case, we will use high temperature gasket maker which is easily available online. The particular product we will use, Permatex Max Temp Gasket Maker, has an intermittent rating for 750°. When applying the RTV silicone, be sure to apply it to both parts and press them firmly against each other, squeezing off the excess. Most RTV's list a cure time between 1 - 24 hours, but in our experience, it is best to wait a couple of

days before testing so there is no uncured silicone.

The nozzle has many key features relating to its internal geometry, including the converging half angle (β), diverging half angle (α), throat area (A_t), exit area (A_e), and the expansion ratio, that determine the efficiency of the nozzle and therefore the performance of the rocket. These parameters can be optimized depending on the expected apogee. In the case of this rocket, we are optimizing for $\frac{2}{3}$ of the expected apogee (3 km), at Las Cruces, New Mexico (1192 m above sea level). Based on this, the nozzle will be optimized for 3192 m above sea level, where the ambient pressure is 10.5 psi.

Here are the final parameters, with their derivations explained below.

$$A_t = 0.620 \text{ in}^2$$

$$A_e = 2.686 \text{ in}^2$$

$$\alpha = 10^\circ$$

$$\beta = 58^\circ$$

5.6.1 Throat Area

The first parameter the team derived is the throat diameter. It is important to optimize this number because a tighter throat helps with engine efficiency, but at a certain point values that are too small can create undesirably high chamber pressures. To arrive at an optimal throat area, the team decided on a design thrust of 1.5 kN, since this was similar to the thrust achieved by teams with the same desired apogee. In a similar fashion, the team decided we did not want to exceed an average chamber pressure of 350 psi, so as to not require thicker, heavier hardware, and to maintain a pressure gradient between the oxidizer tank and the combustion chamber.

The equations that the team used to work toward our throat diameter, are the following:

$$F_t = C_f * A_t * P_c$$

$$A_t = \frac{F_t}{C_f * P_c}$$

$$C_f = \sqrt{\frac{2k^2}{k-1} * \left(\frac{2}{k+1}\right)^{\frac{k+1}{k-1}} * \left(1 - \frac{P_e}{P_c}\right)^{\frac{k-1}{k}} + \frac{P_e - P_{amb}}{P_c} * \frac{A_e}{A_t}}$$

Where:

F_t = Average Thrust (450 lbf)

C_f = Thrust Coefficient

k = Specific Heat Ratio (1.25)

P_e = Exit Pressure (10.1 psi)

P_{amb} = Ambient Pressure (10.1 psi)

P_c = Chamber Pressure (350 psi)

The calculation using these parameters is:

$$C_f = \sqrt{\frac{2 * 1.25^2}{1.25 - 1} * \left(\frac{2}{1.25 + 1}\right)^{\frac{1.25+1}{1.25-1}} * \left(1 - \frac{10.1}{350}\right)^{\frac{1.25-1}{1.25}}} + 0 = 2.075$$

$$A_t = \frac{450}{2.075 * 350} = 0.620 \text{ in}^2$$

These equations were greatly simplified by the fact that we are optimizing the nozzle where the ambient pressure is equivalent to the pressure at the nozzle exit, at $\frac{2}{3}$ of the expected apogee. This takes the last term in the equation for the thrust coefficient to zero, allowing us to use this figure in the second equation. Since the average thrust and chamber pressure are design parameters that the team decided on, we are left with a number for the throat area, which is highlighted above.

To avoid endless, mind numbing algebra, our team wrote a python code that takes the input parameters listed above, and yields the exit area, and exit radius which are necessary for the HRAP simulations and CAD, respectively.

5.6.2 Exit Area

The ratio between the area of the nozzle at the throat (expansion ratio), and the area at the exit, determines how efficiently the nozzle is able to convert exhaust gases into linear velocity, and thus thrust. For relatively low altitude rockets (below 10 km) as this one, "nozzle area ratios are typically between 3 and 30" (Sutton, Biblarz 2017). In the case of this project, the expansion ratio ($\frac{A_e}{A_t}$) ended up being 4.33.³

$$\frac{A_t}{A_e} = \left(\frac{k+1}{2}\right)^{\frac{1}{k-1}} * \left(\frac{P_e}{P_c}\right)^{\frac{1}{k}} * \sqrt{\frac{k+1}{k-1} * \left(1 - \left(\frac{P_e}{P_c}\right)^{\frac{k-1}{k}}\right)}$$

In this equation, the only unknown variable is the exit area, which we can solve for using simple algebra. To simplify the process, this was implemented into the same python program that calculated the throat area. Here is that calculation:

$$\frac{.620}{A_e} = \left(\frac{1.25+1}{2}\right)^{\frac{1}{1.25-1}} * \left(\frac{10.1}{350}\right)^{\frac{1}{1.25}} * \sqrt{\frac{1.25+1}{1.25-1} * \left(1 - \left(\frac{10.1}{350}\right)^{\frac{1.25-1}{1.25}}\right)} = 0.231$$

$$A_e = 2.686 \text{ in}^2$$

From this calculation, we arrived at a value of $A_e = 2.686 \text{ in}^2$. Using the throat area we arrived at earlier, the expansion ratio is approximately 4.33. This value falls in the range we expected from earlier, between 3 - 30.

³The pressure used for this calculation, 350 psi, was the average pressure for the 11 second burn

5.6.3 Converging and Diverging Half Angles

The converging half angle refers to the angle formed between the convergent (inlet) section, and a line parallel to the direction of exhaust gases (typically in the vertical direction). The divergent half angle, similarly, is the angle between the divergent (outlet) section from vertical. This is illustrated in the graphic below, with β , and α representing the convergent and divergent half angles, respectively.

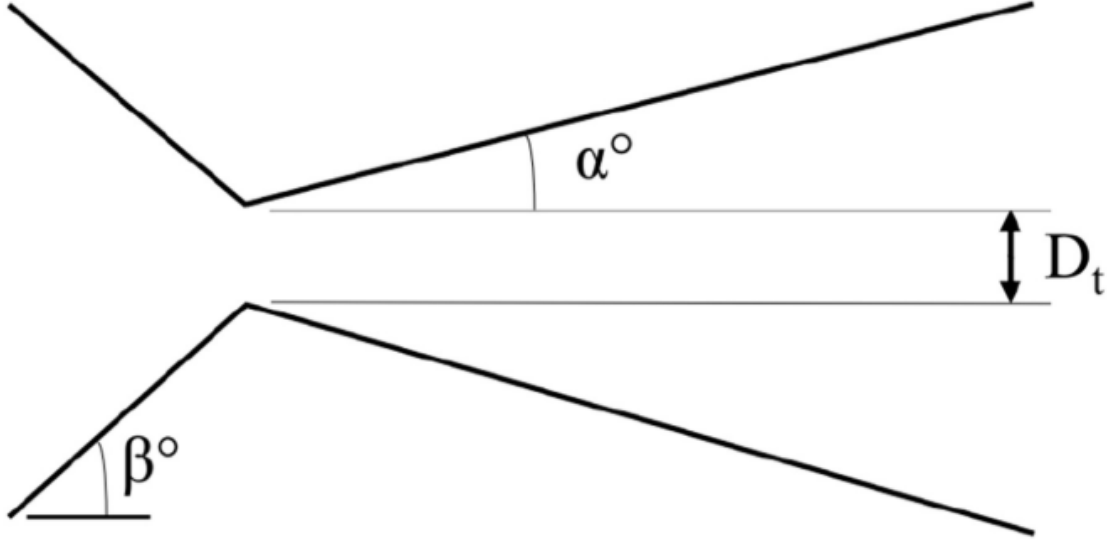


Figure 28: Convergent α and divergent β half angles

Typical values for β range from $45^\circ - 55^\circ$, and values for α are typically in the range of $12^\circ - 18^\circ$. It is important to note, however, that these values are context specific and can vary based on size or mass constraints for your nozzle. In the case of this rocket, we landed on $\beta = 58^\circ$, and $\alpha = 10^\circ$. Notably, both of these values are slightly out of the optimal range, but slight modifications to the values have large impacts on the height of the nozzle, and this configuration gave the most reasonably sized nozzle in terms of mass and height. Further verification was performed on Ansys Flow Simulation to verify that these angles did not result in unnatural or otherwise unexpected internal flows.

To perform the simulation, the average chamber pressure (350 psi), exhaust velocity (1934.0 m/s), and the chamber temperature ($3241^\circ K$) were input as parameters. The average chamber pressure is obtained from the simulation performed in HRAP, the chamber temperature is obtained from inputting our fuel mixture into ProPEP, and the exhaust gas velocity is obtained from the following derivation found in Sutton's Rocket Propulsion Elements:

$$I_t = F_t * t_{burn}$$

$$I_s = \frac{I_t}{m_p * g_0}$$

$$c = I_s * g_0$$

Where:

I_t = total thrust

I_s = specific thrust

m_p = total mass of propellant (8.91 kg)

g_0 = gravity at Earth's surface (9.81 m/s^2)

c = average exhaust gas velocity

From these parameters, we obtain the following equation:

$$c = \frac{1632 * 10.77}{8.91 * 9.81} * 9.81 = 1972.7 \text{ m/s}$$

To find the exit mach number, we use the equations for isentropic gas flow:

$$M_e = \sqrt{\frac{2}{\gamma - 1} * \left(\left(\frac{P_c}{P_a} \right)^{\left(\frac{\gamma - 1}{\gamma} \right)} - 1 \right)}$$

$$M_e = \sqrt{\frac{2}{1.25 - 1} * \left(\left(\frac{350}{10.1} \right)^{\left(\frac{1.25 - 1}{1.25} \right)} - 1 \right)} = 2.87$$

This is the end result, designed in SolidWorks.

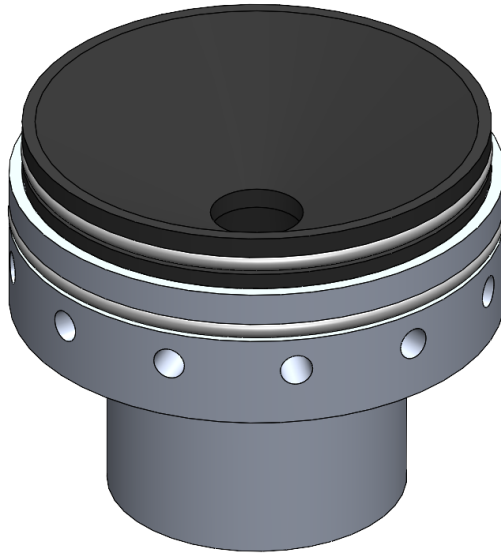


Figure 29: Nozzle Assembly

The nozzle insert shown in the picture above is made of isostatically molded graphite. This material suits our purposes because it has good heat resistance, and it does not erode away easily which

avoids having to re-machine a new part each time a static fire test is conducted. Ideally, the same graphite insert could be used, cleaned, inspected, and reused until the throat has eroded away too much, the part has cracked, or it is otherwise deemed unsuitable for further testing.

The aluminum nozzle carrier's main purpose is, as the name implies, to "carry" the nozzle insert, and structurally connect it with the rest of the motor. To achieve this, there is a single row of twelve 5/16-18 Black Oxide Alloy Steel Bolts that secure it to the case, and an O-Ring groove that seals between the case wall and the nozzle carrier. This is a last resort to prevent any hot exhaust gases from escaping through the case, and heating up the bolts, which would compromise the bolts' structural integrity.

The bolts, which have a threaded length of 0.75 in., will not engage with the case since this would lead to cross threading in one or both components. Since the case is 0.125 in. thick, the length of engagement for the bolts will be 0.625", which is reflected in the CAD.

For reference, some of the relevant dimensions include:

Mass: 1.31 lbs

Height: 3.875 in.

Top Diameter: 4.497"

Bottom Diameter: 2.372"

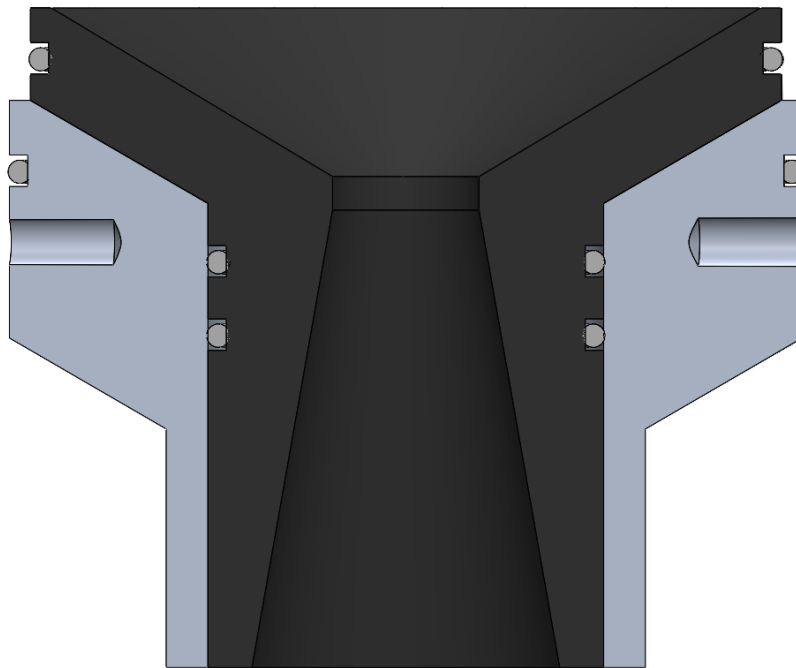


Figure 30: Graphite Insert, Cross Sectional View

The cross section of the graphite insert shown in the picture above reveals the internal geometries

we touched on earlier. The three grooves seen on either side of the nozzle are the O-Ring grooves, the top of which will seal against the ablative liner, and the bottom of which seal against the nozzle carrier. As mentioned earlier, the top seal is necessary to prevent hot exhaust gases from entering between the ablative liner and the case. The two O-Ring grooves between the graphite insert and the nozzle carrier are there in case any exhaust does escape through the top o-ring, it does not exit through the sides of the nozzle, producing unexpected and unpredictable thrust.

The converging and diverging half angles mentioned earlier are more clear in this picture.

5.6.4 Nozzle FEA

Before we test anything, we want to be certain our parts won't crumble, explode, tear, shear, or otherwise fail when we put them under extreme pressure or temperatures. On more complex elements, like the nozzle, the best way to go about this is to use finite element analysis. Our platform of choice is SolidWorks, since our models were designed here and the simulation software offered here works well for relatively simple parts like ours.

Our FEA was configured with the 12 bolt holes on the nozzle carrier as fixed geometries, since we do not expect these to move during tests or fires. The load corresponding to the peak pressure of 653 psi was placed on the inlet face of the graphite insert. The triangular mesh had a maximum size of 0.1 in., and a minimum size of 0.01 in. This yielded the following results:

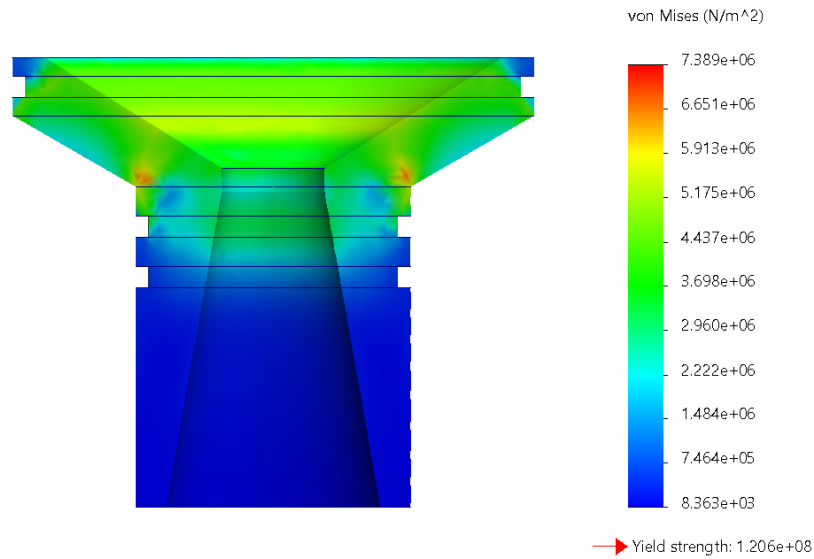


Figure 31: Nozzle Stress FEA

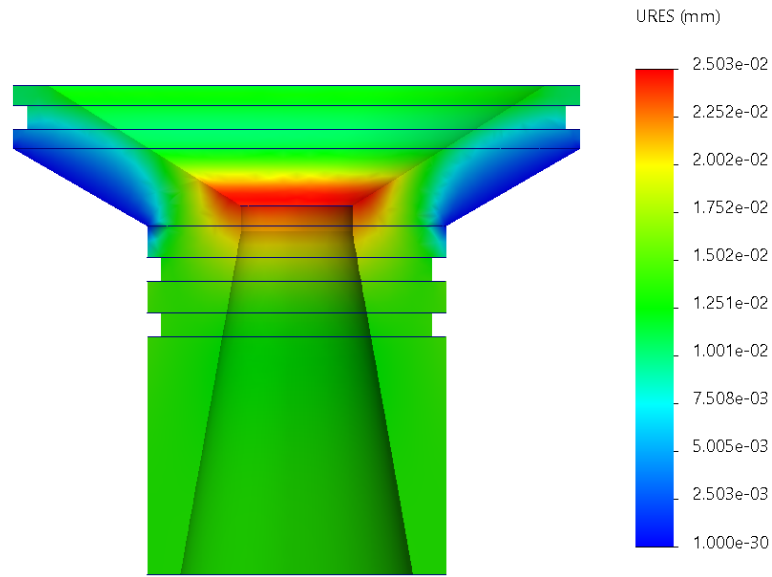


Figure 32: Nozzle Displacement FEA

5.7 Injector

The injector serves the purpose of evenly distributing the Nitrous Oxide oxidizer into the combustion chamber, where it will react with the fuel and heat to produce kinetic energy that can generate thrust. Our design attempts to minimize mass, reduce machining complexity, while still achieving this purpose. The design is shown below.

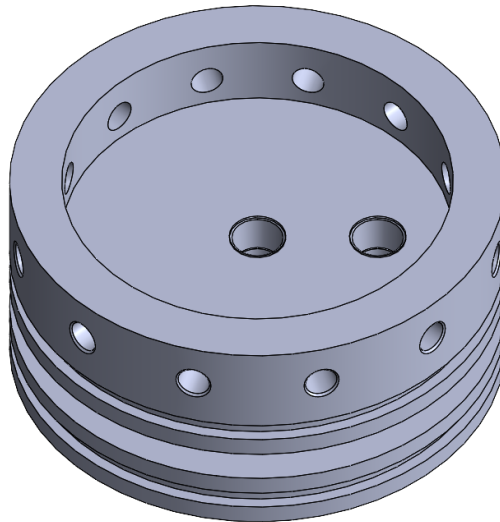


Figure 33: Injector, Isometric View

This view highlights the two NPT tapped holes in the injector. The hole in the center houses the igniter assembly, which produces a flame that gets the combustion process going just as the oxidizer is released from the tank. The hole that is closer to the edge is for a fitting that will connect to the tubing from the Oxidizer tank. It is here that the pressurized Nitrous Oxide will be fed through. This hole is offset by 15° so it does not fall in the same ray as a bolt, which would make the oxidizer hose fitting difficult to thread on and remove.

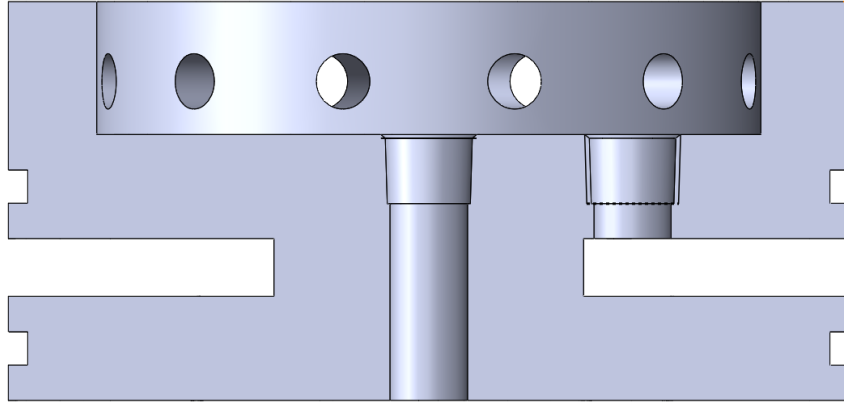


Figure 34: Injector, Cross Sectional View

The cross section gives us a bit more insight into how the igniter works. The large empty space in the center of the part is a chamber sealed by O-rings on top and on the bottom, where the oxidizer (then, a saturated gas) will spread out, and eject through the 8 injector holes. The dimensions of these holes are given in detail in an earlier section. The injector holes are 1.1 in. away from the center of the part, keeping in mind they must be at a smaller distance than the initial port radius (1.2 in.). The machinability of this part may be a problem. We will consult with the machine shop managers to see if the parting tool would be adequate for the job, or if there should be a redesign to split the part in half, so the part is not impossible/dangerous to manufacture.

Up top, there are 12 bolts holding the part to the case. These are using the same 5/16-18 Black Oxide Alloy Steel Bolts from other sections.

5.7.1 Orifice Diameter

The total area of all injector orifices combined can be calculated using the following formula:

$$A = \frac{m_o}{C_d * \sqrt{2 * \Delta P * \rho_o}} = \frac{.654}{.61 * \sqrt{2 * 1.034 * 10^6 * 753.03}}$$

Dividing the output by the intended number of orifices allows you to calculate individual orifice diameters. Typically, a larger number of orifices allows for better atomization of the injected propellant and thus provides higher combustion efficiency. However, it is important to choose an orifice number that corresponds to a machinable orifice diameter. Orifice diameter can be calculated using the following equation:

$$d_{hole} = \sqrt{\frac{4 * A_{total}}{n * \pi}} = \sqrt{\frac{4 * 2.72 * 10^{-5}}{10 * \pi}}$$

Using these equations, we decided to use 10 orifices with an orifice diameter of 1.86 mm. We then rounded the orifice diameter to 1.85 mm as it corresponds to the nearest available drill head size.

5.7.2 Injector Finite Element Analysis

In a similar fashion to the nozzle, we want to be sure that the injector can hold up to the pressure it will experience as the Nitrous Oxide begins to flow through it, and the combustion process begins to occur. The fixtures are again the 12 bolt holes, since we do not expect these to move during test or flight. The load of 960 PSI, that is the pressure from the Oxidizer tank, is placed on the interior faces where the oxidizer feeds in to the injector orifices. a load of 653 psi, from combustion, is placed on the aft end of the injector where the oxidizer exits the orifices.

This yielded the following results in Solidworks Simulation:

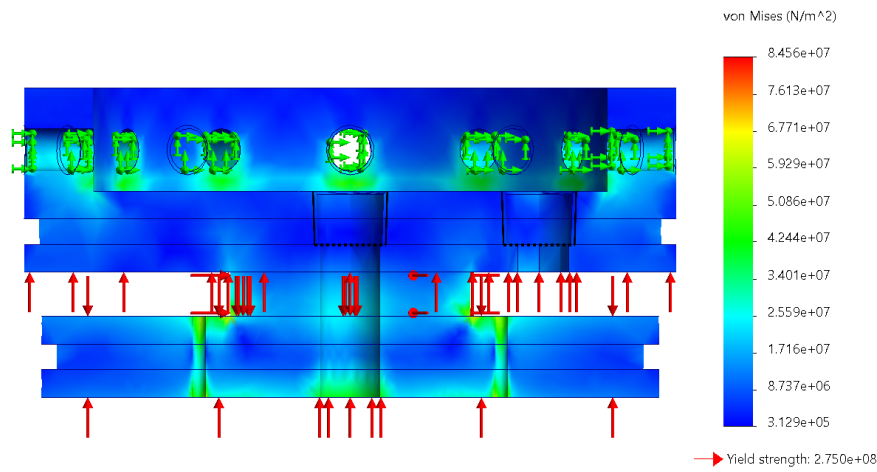


Figure 35: Injector Stress FEA

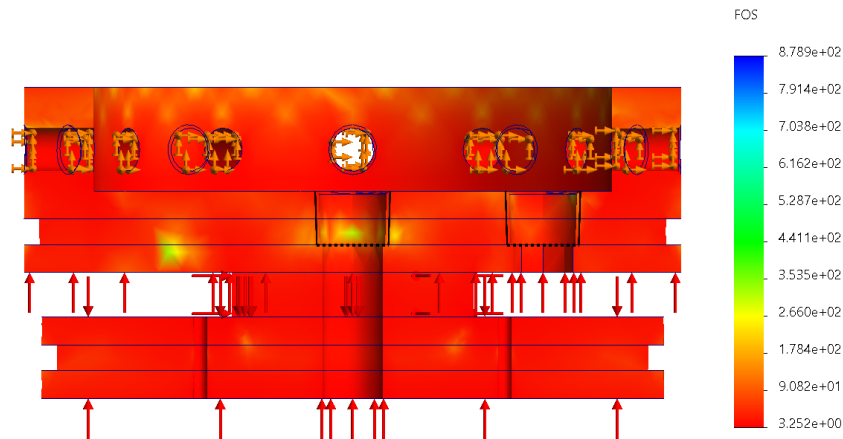


Figure 36: Injector Factor of Safety

5.8 Combustion Chamber Case

The combustion chamber case is the largest, and heaviest part of the rocket. This part houses all the combustion components including the grain, liner, and nozzle. To hold everything together the internal hardware components will be secured to the case using steel alloy bolts. This necessitates pairs of symmetrically spaced bolt holes along the case's circumference, one row at the aft end and the other at the forward end. It is key to size the thickness and position of the bolt holes intelligently and safely, as oversight in either area could be catastrophic for tests or launches.

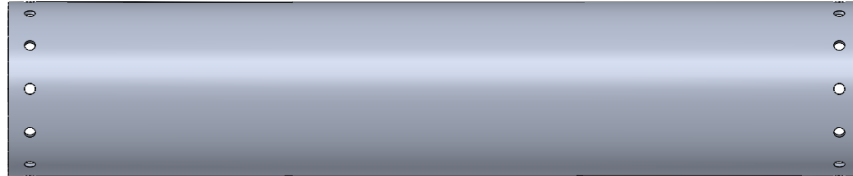


Figure 37: Combustion Chamber Case

There are a few important considerations for the design of a pressure vessel; the wall thickness, the case material, and in our case, the row to edge distance of the bolts. For our purposes, we looked for a material that is easily machinable, relatively low cost, and high strength. We landed on Aluminum 6061, temper 6. This is a common case material, and it should hold up very well for the combustion chamber. Based on the pressures we are expecting, we selected a thickness of 0.125". The distance from the center of a bolt to the edge of the case is 0.625", or around twice the diameter of the bolt. This corresponds to a distance of approximately 0.469" from the edge of the bolt to the edge of the case.

One important note about the combustion chamber case is that it is not the outermost component of the rocket, meaning the fin can will not be situated on it, and it will not experience any aerodynamic forces. Instead, there will be an exterior fiberglass shell that houses the fin can, aero-shells, and related components. The combustion chamber case will be connected to the fiberglass shell by two spacers, one on the forward end of the rocket, and the other on the aft end. These spacers will be constructed out of fiberglass, and will interface with the aluminum case. Of the spacers, one will provide stability so the shell doesn't shift in the vertical direction, and the two together will ensure the shell doesn't move from side to side.



Figure 38: External case view with spacer

The rationale behind not making the combustion chamber the rocket body relates to overall mass, and manufacturing costs (monetary and time-wise). By using a spacer, the combustion chamber hardware components can have a smaller diameter, which is cheaper, lighter, and significantly easier

to manufacture. For a rocket designed to reach an apogee of 3 km, it's not necessary to have a huge combustion chamber.

5.8.1 Combustion Chamber Case FEA

Despite the simplicity of finding safety factors for the case based on the pressure it will experience, our team found it helpful to perform a finite element analysis to check that we got the results we were expecting. The analysis was configured with the 24 bolt holes (12 on the aft, and forward ends) as fixed geometries, and the load placed on the internal face of the combustion chamber case. The triangular mesh had a maximum size of 0.2 in., and a minimum size of 0.05 in.

This yielded the following results:

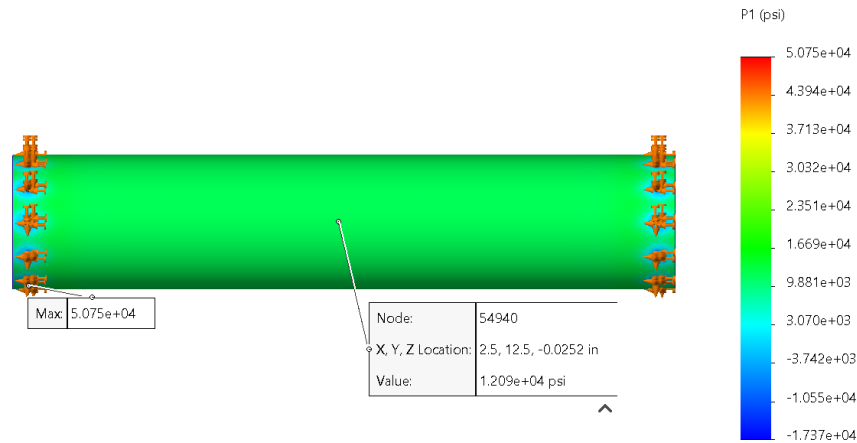


Figure 39: Case Principal (Hoop) Stress

As can be seen in the figure, the majority of the case experiences the same amount of stress, just over 12 ksi. This is consistent with the results from hoop stress calculated, which will be discussed in further detail later.

5.9 Spacers

As mentioned, the rocket body will feature two spacers that connect the combustion chamber with the fiberglass body tube. This component manufactured from Aluminum 6061-T6 for ease of manufacturing, and so the bolts that secure it to the fiberglass are able to thread into it. The spacers serve the purpose of constraining motion circumferentially, and vertically.

Only one of the spacers (on the aft end) will constrain the fiberglass tube in the vertical direction. This spacer is held in place by the same bolts that hold the nozzle in place. Notably, the spacer will not thread into these bolts, it will just be held in place by the head of the bolts, which will fit tightly in the holes drilled out for them. The maximum force experienced by the bolts corresponds to the maximum acceleration of the rocket ($a = 58 \text{ m/s}^2$), the mass of the rocket ($m = 32 \text{ kg}$) distributed evenly along the 12 bolts. Twelve 4-40 screws will tread into this spacer, to hold the

fiberglass in place.

$$F_b = \frac{m * a}{N} = \frac{58 * 32}{12} = 154.67N$$

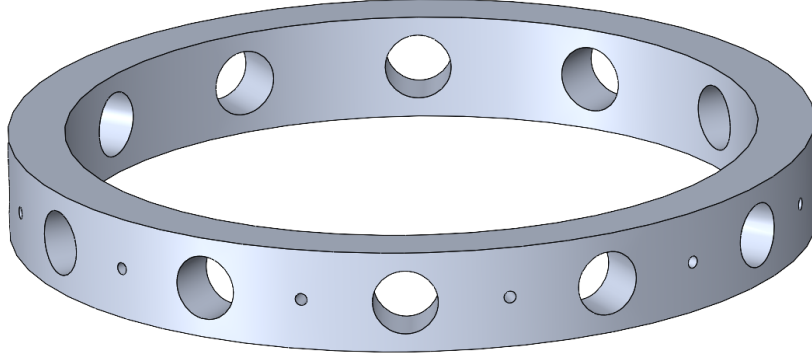


Figure 40: Spacer, Isometric View

The large through holes here will be where the heads of the bolts holding the nozzle in place will rest. The smaller holes are spaced 15° apart from these. These holes are for the 4-40 coarsely threaded screws that hold the fiberglass shell in place.

Both spacers will prevent motion along the circumferential direction. The reason both spacers do not constrain vertical movement is that in order to have two vertically constraining spacers, both would have to be held in place by the bolt heads. Since our case is rotationally symmetric and there is no other reason to perfectly align the aft and forward rows of bolt holes, there is no guarantee that the bolt holes would be perfectly in line with each other. Since the holes on the fiberglass would have to be offset from these by a set amount, it would be unnecessarily difficult to get everything to line up properly, especially since 1 row of bolts is enough to withstand the forces from acceleration.

5.10 Stress Calculations

Stress refers to a force applied over an area. You may notice that all of these stress calculations follow this description, as they are force / area. Thus, the units for stress will always relate to pressure. It is important as an engineer to understand a) where your assembly will experience stress b) how to calculate these stress and ensure your components can hold up to them, and c) understand what these stresses entail. We will do our best to help with this in the following subsections!

5.10.1 Case Hoop Stress

One important consideration in determining the thickness of the case wall for any pressure vessel is the hoop/principal stress that will be experienced by the pressure vessel. Hoop stress is experienced in the circumferential direction from inward pressure within a pressure vessel, and is an important figure to design around. If the hoop stress experienced is greater than the yield stress of the material the pressure vessel is made of, it is guaranteed to fail.

Here is the formula for hoop stress, which we used for these calculations. This is the hoop stress for a thin walled pressure vessel, which is valid because the thickness of our material (.125 in.) is less than 5% of the diameter of the pressure vessel (5 in.) :

$$\sigma_h = \frac{P_c * r_{mean}}{t}$$

Where:

$$r_{mean} = \frac{r_i + r_o}{2} \text{ (2.4375 in.)}$$

t = material thickness (.125 in)

Here is the calculation with the relevant figures:

$$r_{mean} = \frac{2.375 + 2.5}{2} = 2.4375 \text{ in.}$$

$$\sigma_h = \frac{653 * 2.4375}{0.125} = 12,734 \text{ psi}$$

From this calculation, we obtain a value of hoop stress $\sigma_h = 12,734 \text{ psi}$. The combustion chamber case, which will be constructed of Aluminum 6061-T6, has a yield stress equivalent to 35,000 psi. To find the factor of safety, we divide this number by the hoop stress we just calculated. Here is that calculation:

$$SF = \frac{35,000}{12,734} = 2.74$$

This safety factor exceeds the factor of 2, so we do not expect this to be a failure mode.

5.10.2 Case Axial Stress

Another important consideration for pressure vessels is the axial, or longitudinal stress. This stress is experienced along the length of the pressure vessel, when the forces tend to change the length of a body. Our team used the following formula to calculate the axial stress:

$$\sigma_a = \frac{P_c * (D - t)}{4 * t}$$

Where:

D = Case Outer Diameter (5 in.)

Utilizing the parameters, we get the following equation:

$$\sigma_a = \frac{653 * (5 - 0.125)}{4 * 0.125} = 6,367 \text{ psi}$$

Given the yield strength of Aluminum 6061-T6, here is the calculation for the safety factor:

$$SF = \frac{35,000}{6,367} = 5.50$$

5.10.3 Case Tensile Stress

A potential failure point for our case is in the space between the bolt holes. We are assuming the entire case will be pressurized (although this is hopefully not the case), so it is important to consider the stretching of the case material circumferentially from the internal pressure. This is typically only a problem for motors where there are large quantities of bolts in a single row, or where there are very large bolt holes (for example, with countersunk bolts). To verify that we won't experience any problems, the stress calculation is shown below:

$$\sigma_{tensile} = \frac{\frac{\pi}{4} * D_{case,inner}^2 * P_{max}}{((D_{case,outer} - t) * \pi - (N * d_{bolt,major})) * t}$$

$$\sigma_{tensile} = \frac{\frac{\pi}{4} * 4.75^2 * 653}{((5 - 0.125) * \pi - (12 * 0.313)) * 0.125} = 8,008.4psi$$

$$SF = \frac{35,000}{8,008.4} = 4.37$$

The factor of safety is well above the required minimum ($SF = 2$), so we do not expect failure in this area.

In accordance with our demonstrated love for Python, we wrote a python program that calculated these two stresses, and used these figures to calculate Von Mises stress and from there, predict failure points and determine a safety factor based off the pressures we are expecting.

The formula for Von Mises stress is the following:

$$\sigma_v = \sqrt{\sigma_a^2 + \sigma_h^2 - \sigma_a * \sigma_h}$$

With the values for hoop and axial stress we calculated above, we can find the Von Mises strength, and determine the overall safety factor of the case. Here is that calculation:

$$\sigma_v = \sqrt{6,367^2 + 12,734^2 - 6,367 * 12,734} = 11,028 \text{ psi}$$

Here is the formula and calculation for the safety factor using Von Mises:

$$SF = \frac{\sigma_{yield}}{0.5 * \sigma_v} = \frac{35,000}{0.5 * 11,028} = 3.17$$

5.10.4 Bearing Stress Calculation

Another important consideration is the bearing stress, which is stress applied perpendicular to a materials surface, causing it to deform. In our case this is from the bolts onto the case material. Notably, bearing stress is not compressive stress. Rather, it is an internal stress caused by compressive forces. The formula to calculate this figure is:

$$\sigma_{bearing} = \frac{F_b}{d_{major} * t} = \frac{964.29}{0.313 * .125} = 24,646.5 \text{ psi}$$

$$SF = \frac{\sigma_{bearing,Al}}{\sigma_{bearing}} = \frac{56,000}{24,646.5} = 2.27$$

This formula utilizes the bearing strength of Aluminum 6061-T6, not the yield strength. The calculation demonstrates our factor of safety above 2, so we are in the clear here.

5.10.5 Tear Out Stress Calculation

Another common failure point related case is shear tear out stress, which is similar but not identical to tear out stress. This is a type of stress where, rather than the bolt itself shearing, the material surrounding the bolt hole will itself shear. In this case, the material of concern is the combustion chamber case.

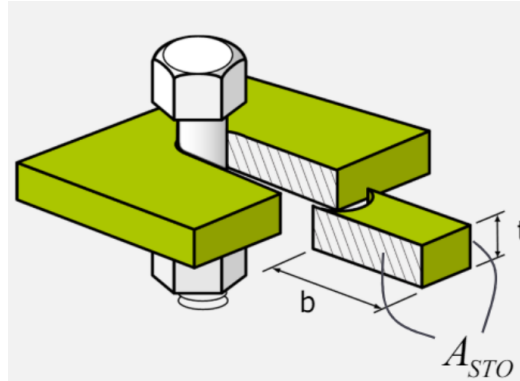


Figure 41: Relevant shear tear out area, A_{sto}

To perform the calculation for shear tear out stress, the force experienced by an individual bolt is multiplied by the cross sectional area of the case material A_{sto} as shown in the diagram. For our model, the center of the bolt is 5/8" (.625) away from the edge of the case. Subtracting the radius of the bolt $\frac{.313}{2}$, we find the bolt to edge distance, $d = 0.469$ ". Earlier, we selected the case thickness, $t = 0.125$ ". Using the figure for F_b we calculated earlier, we perform the following calculation:

$$\sigma_{sto} = \frac{F_b}{2 * A_{sto}} = \frac{F_b}{2 * d * t}$$

$$\sigma_{sto} = \frac{964.29}{2 * 0.469 * 0.125} = 8224.2 \text{ psi}$$

$$SF = \frac{0.577 * 35,000}{8224.2} = 2.45$$

Where:

d = Bolt to edge distance (.469 in.)

t = Material thickness (.125 in.)

5.10.6 Bolt Shear Stress Calculations

The most likely failure point for the bolts the hybrid engine will be using is in shear. Shear failure refers to failure from shear stress, that is, from forces that are in the same plane as the cross section of the fastener material. Below is a pictorial representation to demonstrate this. In our case, the force would come from the internal pressure pushing against the nozzle that transfers onto the nozzle carrier, which is counteracted by the normal force from the case.

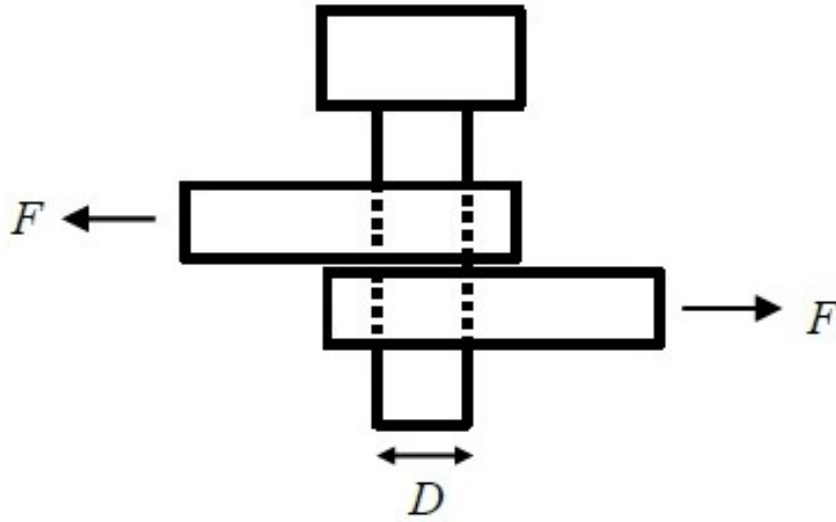


Figure 42: Shear Stress Representation

The formula to calculate shear stress divides the Force experienced by each bolt by the effective cross sectional area of that bolt. By "effective" we are referring to the area not including the threads. Here is that as a formula:

$$\sigma_s = \frac{4 * F_{bolt}}{\pi * d_{bolt}^2}$$

To find the force each bolt would experience F_b , it is necessary to first multiply the area on the interior of the combustion chamber by the maximum internal pressure. From here, divide by N bolts, since ideally, the force would be evenly distributed among the set of bolts. Here is that calculation for our combustion chamber.

$$F_{bolt} = \frac{P_{max} * \pi * r_i^2}{N} = \frac{653 * \pi * 2.375^2}{12} = 964.29 \text{ lbf}$$

The bolts we are using are 5/16-18 Black Oxide Alloy Steel bolts, which have an effective diameter of .313 in. Thus, the shear stress for the bolts using the formula from above is:

$$\sigma_s = \frac{4 * 964.29}{\pi * .252^2} = 19,333.8 \text{ psi}$$

The bolts we selected have a yield strength of approximately 90,000 psi, according to the product specifications on McMaster Carr. A common approximation for the conversion from yield strength to shear strength σ_{bolt} is multiplication by a factor of 0.55 to 0.6. for our purposes, we will go right in the middle with a conversion factor of .575, for an approximate shear strength of 51750 psi. To find the factor of safety, divide the shear strength of the bolt by the force experienced by the bolt, as follows:

$$SF = \frac{\sigma_{bolt}}{\sigma_s} = \frac{51,750.0}{19,333.8} = 2.69$$

This factor of safety is well above the required factor of safety of 2, so we do not expect a failure of the bolts in the shear direction.

5.11 O-Ring Selection

O-Rings are a small, yet critical component to any rocket motor. In any interface between two materials, there will always be imperfections that will make it so there is not a true seal between them. Such is the case between our injector, nozzle assembly, and the case. O-rings are typically made of silicone, nitrile, or other rubbers, and are great tools to address this problem because they can compress and seal off the imperfections, helping create a pressure tight seal. O-rings can be used in face gland seals, or male - female gland seals. Our rocket will use male - female gland seals.

It is important to reference appropriate materials, like the Parker O-Ring Guide, when deciding the o-ring pressure vessels. These reference materials provide accurate dimensions for the dimensions of the O-Ring grooves, based on the interior dimension of the pressure vessel they will be sealing. In our case, we had three major interfaces that we needed to seal against; nozzle carrier - case (1), graphite insert - nozzle carrier (2), graphite insert - ablative liner (3). The table below contains the relevant dimensions of each.

Dimension	1 (4.75" ID)	2 (4.5" ID)	3 (2.375" ID)	Tolerances
O-Ring Series	2-046	2-044	2-027	N/A
Groove Depth	4.528 "	4.278 "	2.153 "	+ .000 "
Plug Diameter	4.747 "	4.497 "	2.372 "	+ .000 .001 "
Groove Width	0.187 "	0.187 "	0.187 "	+ .005 "

The O-Rings used for this motor belong to the 200 series, which have a cross sectional area of 0.125 in. The team decided against using 300 series O-Rings despite the lower tolerances because in previous projects, the large O-rings have made integration, and de-integration, excessively challenging, and the large O-Rings often sliced as they passed through the bolt holes since they expand more than 200 series O-Rings

As an additional safeguard to avoid slicing the o-rings and to ensure the o-rings can operate as needed, proper lubrication is needed. Within the combustion chamber, there are areas of high and low temperatures, with a different lubrication method needed for each. For lubricating the o-rings on the aft end, the team will use Molybdenum lithium-graphite multipurpose grease. On the forward end, the team will use Krytox, a low temperature grease.

6 Simulation

6.1 Hybrid Rocket Analysis Program

In order to make predictions on the performance of our engine, we used the program HRAP to generate simulated thrust curves. While the program was instrumental in the design of our engine, it was at times difficult to work with due to the seemingly circular design cycle. For example, designing the fuel grain requires certain assumptions to the regression rate and mass flow rates in our system. However, the program would give these design parameters as outputs that were different from what we had originally used. For that reason, we had to ensure that whatever outputs that were generated in the simulations aligned with the numbers we designed with.

6.1.1 HRAP Inputs

The following shows our configuration of HRAP to achieve our desired thrust and performance requirements. All inputs reflect the design decisions explained in this document:

Hybrid Rocket Analysis Program

Simulation Configuration
Results
About

Configuration Name
CDR_Hybrid

Tank Dimensions

Oxidizer Tank Volume
6.25 L

☐ Calculate Oxidizer Tank Volume By Dimensions

Oxidizer Tank Length
0 cm

Oxidizer Tank Diameter
4.5 in

Combustion Chamber Volume
0 cm³

☒ Calculate Combustion Chamber Volume By Grain Dimensions

Nozzle Configuration

Nozzle Throat Diameter
0.88 in

Nozzle Expansion Ratio
4.33

Nozzle Efficiency
97 %

Nozzle Discharge Coefficient
0.95

Mass Properties

Oxidizer Tank Location
72.55 in

Fuel Grain Location
91.64 in

Empty Motor Center of Mass
52.33 in

Empty Motor Mass
9.478 kg

☐ Calculate Mass Properties

Initial Conditions

Starting Tank Temperature
297 K

Starting Chamber Pressure
1 atm

Tank Fill Percentage
95 %

Ambient Pressure
1 atm

Propellant Configuration

Load Propellant Configuration

Propellant Name
Paraffin

Propellant Density
900 kg/m³

Regression Coefficient (a)
0.1435 mm/s

Regression Exponent (n)
0.5275 kg/m²-s

Length Exponent (m)
0 m

Constant OF Ratio
8.27

C* Efficiency
100 %

Grain ID
3 in

Grain OD
4.25 in

Grain Length
11.73 in

Injector Configuration

Injector Diameter
1.85 mm

Injector Discharge Coefficient
0.61

Number of Injectors
10

Vent State
None

Vent Diameter
0.028 in

Vent Discharge Coefficient
0.75

Simulation Configuration

Max Simulation Run Time
8.5 s

Max Burn Time
0 s

Simulation Timestep
0.001 s

Regression Model
Shifting OF

Load Simulation Configuration
Run Simulation
Save Simulation Configuration

Figure 43: HRAP Configuration Screen

6.1.2 HRAP Outputs

The following image shows the results of the simulation after configuring the program as described above. The graph shown depicts thrust over time:

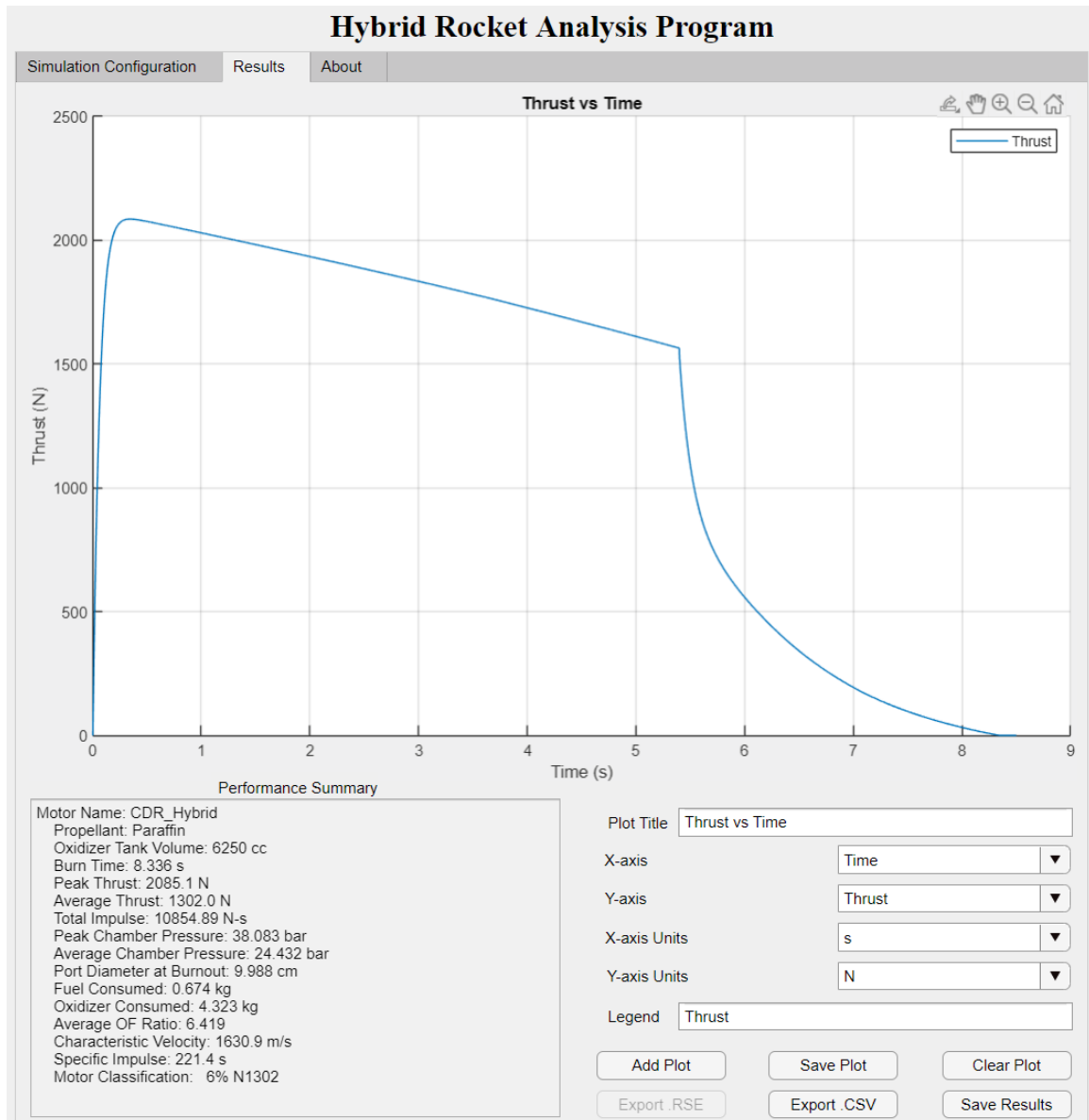


Figure 44: HRAP Results Screen

6.2 OpenRocket

Using the thrust curves generated from HRAP, we imported those thrust curves into OpenRocket to get an estimate of a rocket's apogee that were to use our hybrid engine. While there is no finalized design for the rocket that this hybrid engine will be incorporated into, we can still estimate the apogee by making predictions of the rocket's design. To get an estimated apogee, we created a rocket with the location and masses of pivotal components, while also ensuring stability by putting the center of pressure below the center of gravity. The image below depicts an prediction of what

the final rocket may look like:

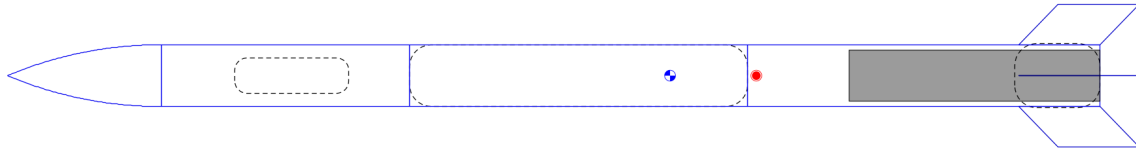


Figure 45: Rocket Prediction

In our OpenRocket configuration, we have a 13.85 kg mass located in the middle of the rocket. This corresponds to the full weight of the oxidizer feed system with the mass of oxidizer included. Below that is the combustion chamber assembly, housed entirely in the dark gray region. This weight is about 7.5 kg and includes the full weight of the paraffin wax. A 4 kg weight is placed near the top of the rocket and is an estimated placement and mass for any necessary parachute and electronics. A 1 kg weight is placed on the outer diameter of the combustion chamber and corresponds to the fin can. Using this rocket configuration and the simulated thrust data, we are able to estimate our apogee:

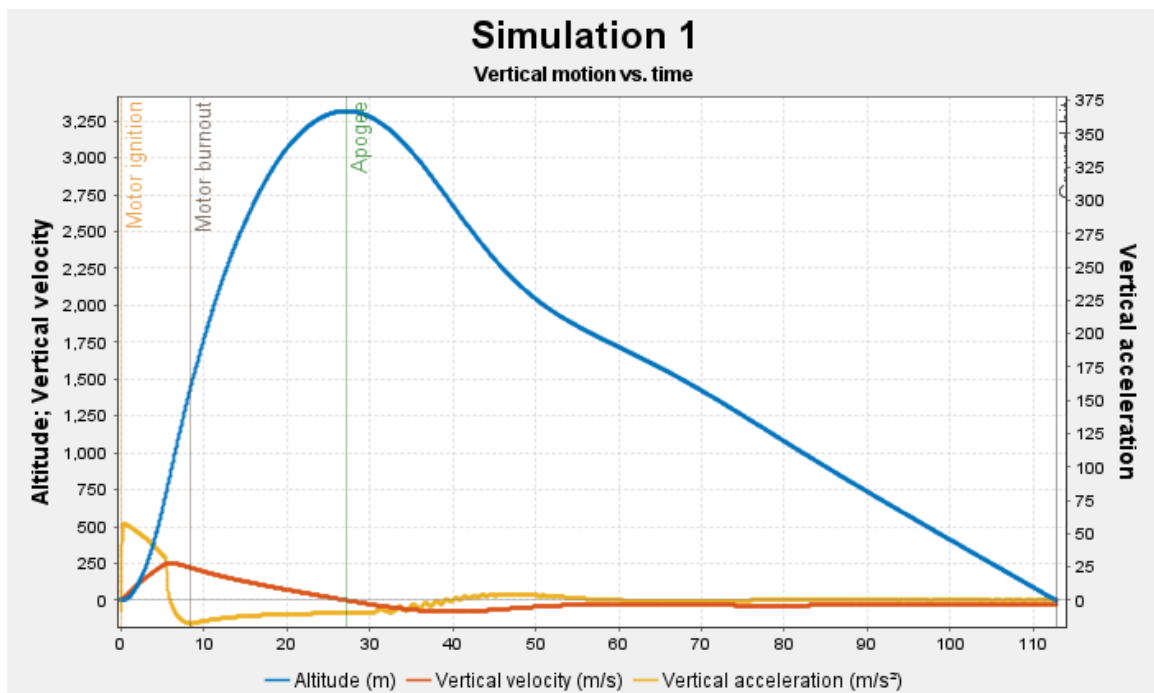


Figure 46: Apogee Simulation

As demonstrated in the altitude simulation, we have an estimated apogee of 3.318 km. This roughly corresponds to our desired apogee of 10,000 ft with an added 10 percent buffer.

7 Testing

7.1 Test Stand

A static fire is one of the most rewarding experiences you get working in experimental rocketry. Unfortunately (but unsurprisingly) it is also one of the most dangerous. This is where all the design, manufacturing, and integration culminates, in what is hopefully one big plume. In order to test safely, you need to mechanically constrain your motor so that once it does ignite, it stays still. Hence, a "static" fire. This is achieved by using a test stand, which is typically a rigid steel, or aluminum structure that can test either vertically or horizontally. Testing horizontally is easier if you have a fixed platform that you can mount the test stand to. It is also simpler for hybrids because you don't need to interface between the oxidizer tank and the combustion chamber. Vertical testing is easy if you have a solid motor, since you don't have the added complexity of an oxidizer tank and firing straight down means you only need a flat, hard ground to push against. Since the team was not able to acquire reasonably priced fiberglass in the time frame we needed, and so could not manufacture the interface between the oxidizer tank and combustion chamber, we went with the horizontal testing option. The CAD is shown below:

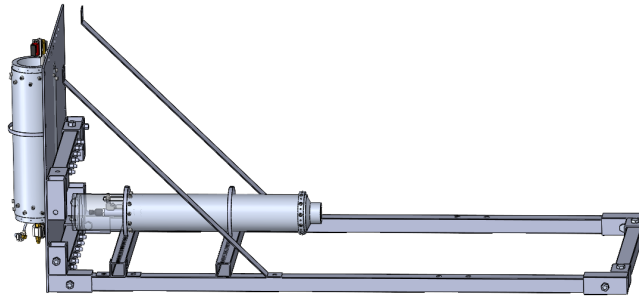


Figure 47: Test Stand Assembly

Another important note for horizontal testing is that since we do not want to compress the fittings, igniter, hoses, etc., we need to make a cap that protects these and transmits thrust. This part is explained in further detail in the next subsection.

Given that we are testing horizontally, we need a way to constrain the test stand's movement. Ideally, it would be bolted to a concrete pad at our testing location. Unfortunately, we don't have permission to do this (we also don't have the time, or resources). Instead, we are driving metal stakes into the ground that prevent it from sliding as thrust is produced. Additional stakes will be attached with ratchet straps to ensure the test stand is rigidly held in place. If the testing location permits, we also plan on propping the test stand against a wall so it doesn't slide along the ground.

Besides holding the motor in place, the test stand must also allow for pressure and thrust data to be collected. This data is measured by a pressure transducer and a load cell. The latter of these is mounted directly on the test stand (directly against the testing cap on the forward end of the motor). The former is screwed into the injector. Data from these sensors is transmitted through electrical connections that are wired to a DAQ (data acquisition system) that receives and reads the data.

7.2 Combustion Chamber Cap

In order to effectively collect thrust data during out static fire, a flat surface will be needed between the load cell located on the test stand and the engine itself. Considering that our combustion chamber assembly will be what's pushing against our load cell, it was necessary to cover the instrumentation on the top of the combustion chamber assembly with some sort of cap.

In order to simplify the cap as much as possible, it will be machined out of extra 6 in. OD aluminum stock. The cap will not require any fasteners, instead the pieces will be tightly toleranced and held together through friction. The top piece is machined from extra 6" aluminum bar left over from the injector and bulkheads. The case is machined from extra 6" OD tube that is 1/8" thick.

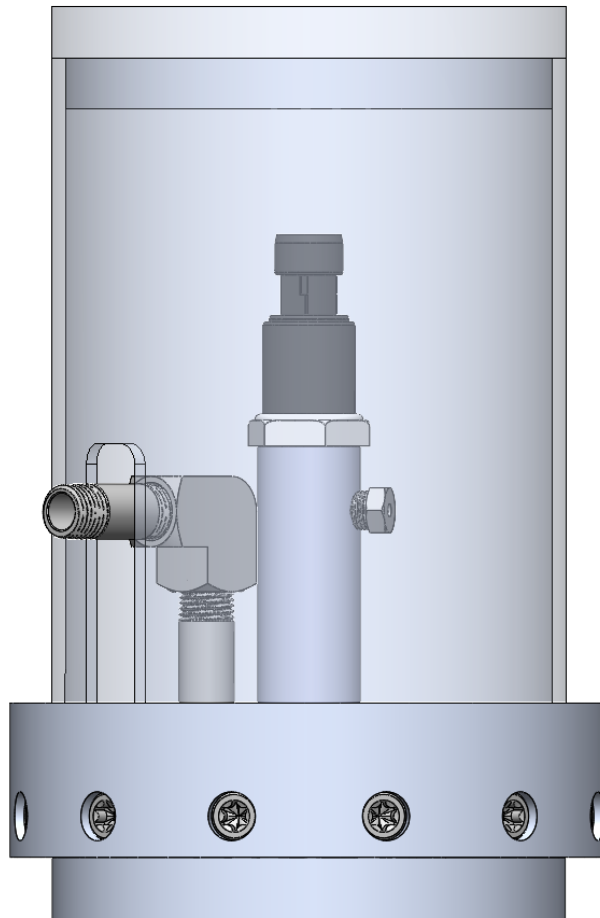


Figure 48: Combustion Chamber Cap

7.3 Remote Control Firing System

To safely conduct all testing of our hybrid engine, we need some way to control our valves and igniter remotely. During testing, personnel should be 200 ft away from the test stand, meaning we not only

need something that can control our systems remotely, but from a long distance away as well. The solution our team chose is an RC controller normally used to control small planes and cars from a long distance.

The entire remote control firing system will consist of an RC transmitter controller, a LiPo battery, voltage converter, electronic relay, and receiver. The battery will be used to power the servo motors used in our servo-ball valve assemblies, as well as our DAQ and igniter relay. In order to supply the necessary voltage for these components, a 12-5V converter is used to reduce the voltage coming from the battery. In order to remotely control our servo motors and igniter relay, a receiver will be placed between the battery and these components. This will allow us to control our valves and igniter using the RC controller from the necessary distance away.

7.4 Instrumentation

7.4.1 Load Cell

The load cell is a type of sensor that transforms mechanical loads into readable electronic data that can be used to accurately measure an applied force. In our case, we will be using a button load cell, which works by using strain gauges that measure mechanical deformation and record it as an electrical response. The voltage produced by load cells is often very low, so to get usable readings, a voltage amplifier is used. This transforms the scale of readings from millivolts to volts.

Selecting the right load cell is important to get data with a high level of fidelity, and as little noise as possible. Some factors that go into this selection are the way the load will be applied, and the range of values you are expecting. In the case of these tests, we will apply a load using an aluminum coupler that exerts a vertical force on the load cell. For this reason, we found a button load cell to be most useful. Our tests are expected to be in the ballpark of 0-2 kN, which is around 1/4 the capacity of the 1T (8.9 kN) load cell we are using.

The specific load cell we will be using is manufactured by CALT, part number DYM102/105, and is available on Amazon. It is rated for weights between 20 kg - 10T.



Figure 49: Load Cell

This sensor does not directly give you the readings you would line, since it only produces voltage readings. To give these readings some meaning, calibration is required. This is typically performed using either an instron or objects of known weights. Each weight/load should be placed on the load cell in order of ascending weight/load, and held for a couple of seconds. This will show up as a plateau on the voltage readings, which can be used to find a line of best fit. With this data, you can make it so your data directly translates and you can read the load directly after a test.

7.4.2 Pressure Transducer

A pressure transducer is a type of sensor that produces an electrical signal as a response to deformation in the transducer's diaphragm. The diaphragm is an internal components that bends with pressurized fluids. This can be connected directly to a data reading device, the DAQ, with no need for a voltage amplifier. Good practice when looking for a suitable pressure transducer is to have the transducer's operating pressure close to the pressure you will be reading. For our specific sensor, the operating pressure is 500 psi, which is perfect for the application of the combustion chamber. Another important consideration is the transducer's over pressure, which is the maximum pressure that can be applied before the transducer is out of calibration. For us, that number is 1740 psi. This number is above 1.5 x the maximum operating pressure of both the combustion and oxidizer chamber's, which is the pressure at which we will be conducting the hydrostatic tests. The transducer has a burst pressure of 3000 psi, but we plan to stay well below this. Another important consideration is the thread size. All pressure transducer's should use an NPT male thread, since this is ideal for a pressure seal. The exact size, however, varies between transducers. For the product we are purchasing, the hole is 1/8" NPT.

The specific pressure transducer we will be using is manufactured by Honeywell Sensing and Productivity Solutions, part number MIPAN2XX500PSAAX. This sensor is available on Digikey.



Figure 50: Pressure Transducer

As mentioned, the pressure transducer gives us information in the form of voltage readings. We can not directly interpret these voltage readings, however, without first performing a calibration. This

is a pretty simple process but it must be performed carefully, as your pressure readings will only be as accurate as the calibration you make. The process is simply to have your hydrostatic test set up, and record the data in known intervals that you can interpret later. For example, you can test in six intervals of 250 psi, holding the pressure at each point. This will show up as 6 distinct plateaus in your voltage readings. Since the data is likely in a excel spreadsheet, you can find a line of best fit that can be applied to your pressure voltage data to directly spit out pressure readings. Identical pressure transducers (same model) should have very similar calibrations, but it is always better to play it safe and calibrate them individually.

7.5 Testing & Integration Procedures

7.5.1 Igniter Plug Integration Steps

1. Using latex gloves, apply acetone to a clean paper towel and carefully wipe down the 1/8" NPT plug.
Be sure to clean the inside tapered surface and the exterior threads.
2. Grab an e-match from the tail, and cut it so you have around 5" of wire to work with.
3. Split the e-match at the tail end, and strip the wires to reveal around 1" of exposed copper wire on the positive and negative ends.
4. Pass the e-match through the tapered holes in the 1/8" NPT plug, making sure that the head of the e-match points in the direction of the threads, i.e. it will be nearest the 1/4" hole, and the wires will run out of the smallest hole. Apply 2-3 layers of electrical tape to the threads, with the tape facing outward.
5. Prepare a small amount of the epoxy mixture in a paper cup, that is around 1.25 x what you predict the hole will need.
6. It is always better to mix a little more than you need and not have to fall short.
7. Cover the tip of the e-match in a thin layer of tape Keeping the head of the e-match just outside the threaded end of the plug

7.5.2 Injector Integration Steps

1. Using latex gloves, apply acetone to a clean paper towel.
2. Wipe all interior and exterior surfaces of the injector, igniter cartridge, and the male/female threads on the pressure transducer and steel e-match plug.
3. Using a fine point like an exacto knife or thin flathead screwdriver, gently remove residual teflon tape from these threads if there is any, before you wipe the part down.
4. Apply teflon tape to the e-match plug, both ends of the steel fitting, and the pressure transducer. The tape should go on in the clockwise direction (from your perspective looking at the threads face on) of the male thread.
5. Apply the tape along the entirety of the threads. Tightly wrap 3 layers of teflon tape around each thread.
6. Adjust the width of the adjustable jaw to fit tightly onto the hex head of the e-match steel plug.
7. Verifying the male and female threads are free of obstructions, hand tighten the steel plug onto the hole on the side of the igniter cartridge. Watch the teflon, if it begins to bunch up, stop, remove the tape, clean the threads, and try again.
9. Once the plug is hand tight, use the adjustable wrench to drive it as far as it can reasonably go. Again, verify the threads haven't bunched up. If they have, deintegrate and try again.
10. Adjust the width of the adjustable jaws on the wrench to fit tightly on the hex drive of the

pressure transducer. Verifying the male and female threads are free of obstructions, hand tighten the pressure transducer onto the hole on the end of the igniter cartridge closest to the e-match plug.

11. Once the plug is hand tight, use the adjustable wrench to drive it as far as it can reasonably go. If you notice the teflon tape bunching up, unscrew, remove it, and try again.
12. Verifying the male and female threads are free of obstructions, hand tighten either end of the steel NPT fitting onto the hole on the hole in the center of the injector.
13. Using the pipe wrench, drive the steel fitting as far as it can reasonably go. If you notice the teflon tape bunching up, stop, clean it up, and try again.
14. Insert the Estes “A” motor through the free end of the igniter cartridge, and push it until the top sits around where the tip of the e-match is.
15. Verifying the male and female threads are free of obstructions, hand tighten the bottom end of the igniter cartridge onto the other end of the steel NPT fitting. Tighten using the pipe wrench.

7.5.3 Nozzle Integration Steps

1. Using latex gloves, apply acetone to a clean paper towel.
2. Thoroughly clean the interior and exterior surfaces of the nozzle carrier and insert. Be sure to remove all visible grime.
3. Test fit the graphite and nozzle carrier interface. There should be a snug fit that is not too hard to separate.
4. Lubricate two 2-027 O-rings using the high temperature grease, and place them on a clean paper towel.
5. Insert the O-rings through the outlet side of the nozzle, until they sit in the grooves. Replace your gloves.
6. Apply a dab of RTV silicone to the tips of your fingers, and apply this to the bottom angled face of the graphite insert, and the top angled face of the nozzle carrier RTV should thoroughly coat these surfaces in their entirety, no aluminum or graphite should be visible in these areas.
7. Press the nozzle carrier and insert together tightly.
8. Some RTV should squeeze out, wipe away this excess.
9. With the outlet side down, and now that all excess RTV has been removed, place the nozzle flat on the table. Place a weight, like aluminum round bar stock, on top of the graphite.
10. This assembly will need to dry for 5 days. Label the time you finished assemblage, and the time it will be done, with instructions to not move it until then.

7.6 Hydrostatic Testing

A hydrostatic test is a real world verification that the finite element analysis, and hand calculations we conducted throughout this report are accurate. The spaceport requirement for hardware is that it is hydrostatic tested to 150% of the maximum expected pressure, for at least 2 x the expected burn time. This corresponds to around 1500 psi for the oxidizer tank, and 1000 psi for the combustion chamber. Although this rocket is not necessarily for spaceport, we are designing it as if it were for a spaceport rocket, so we are imposing this requirement as well.



Figure 51: Hydrostatic Test Pump

A pump like the one shown above is needed for this procedure. This is just a random hydrostatic test pump found on Google images, to demonstrate what they might look like. You have the option of using a manual pump or a gas/electric pump. The latter option is physically easier to use, often have higher pressure ratings, but are heavy and very, very expensive. For this reason, we have historically tested using a manual pump rated for 2 ksi. Theoretically, this is also something you could construct from scratch, though that is easier said than done (but definitely possible).

In case you are wondering why we perform a hydrostatic test (as opposed to just pressurizing the vessel with air) the reasons have to do with the distinct properties of air and water. Water is a nearly in-compressible fluid, and it is not affected nearly as much by temperature as air is. Air, on the other hand, is very compressible and its density varies heavily with temperature. Since air is compressible, it will be in an unnatural state when pressurized, and try to push back on the pressure vessel in order to relieve pressure. This can lead to an explosion, if the pressure vessel bursts. Water, on the other hand, would simply leak out, and would not cause a dramatic explosion. So, for safety reasons, we test pressure vessels with water. This is also the reason you must make sure there are no air bubbles present in the pressure vessel or the pump you are using to pressurize it.

7.6.1 Igniter Test

7.6.2 Hotfire Testing

7.7 Sources